

Accident Reconstruction and ADAS

Alan Moore, P.E., ACTAR

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Alan Moore, P.E.

- Mr. Moore is a mechanical engineer and principal of A.B.Moore Forensic Engineering. He specializes in vehicle accident reconstruction, vehicle design analysis, and mechanical engineering consulting.

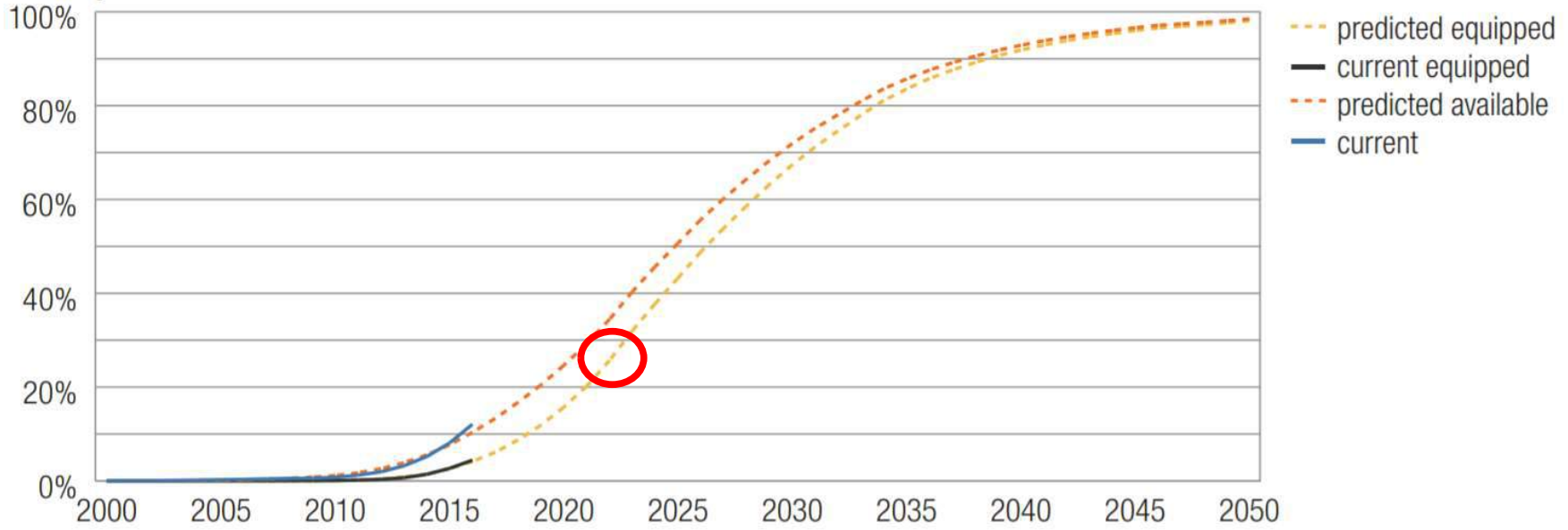
- Mr. Moore holds a Bachelor of Science in Mechanical Engineering from Michigan State University and a Masters degree in Business Administration from the University of Florida. He is a licensed Professional Engineer, a Board Certified Forensic Engineer, and an ACTAR-certified accident reconstructionist. His past experience includes two decades of accident reconstruction and automotive engineering. He previously worked at Ford Motor Company as a design engineer on the Ford Expedition and Excursion. Mr. Moore also serves as a high-performance driving coach for aspiring race car drivers through the Porsche Club of America.



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The Future of ADAS and Accident Reconstruction

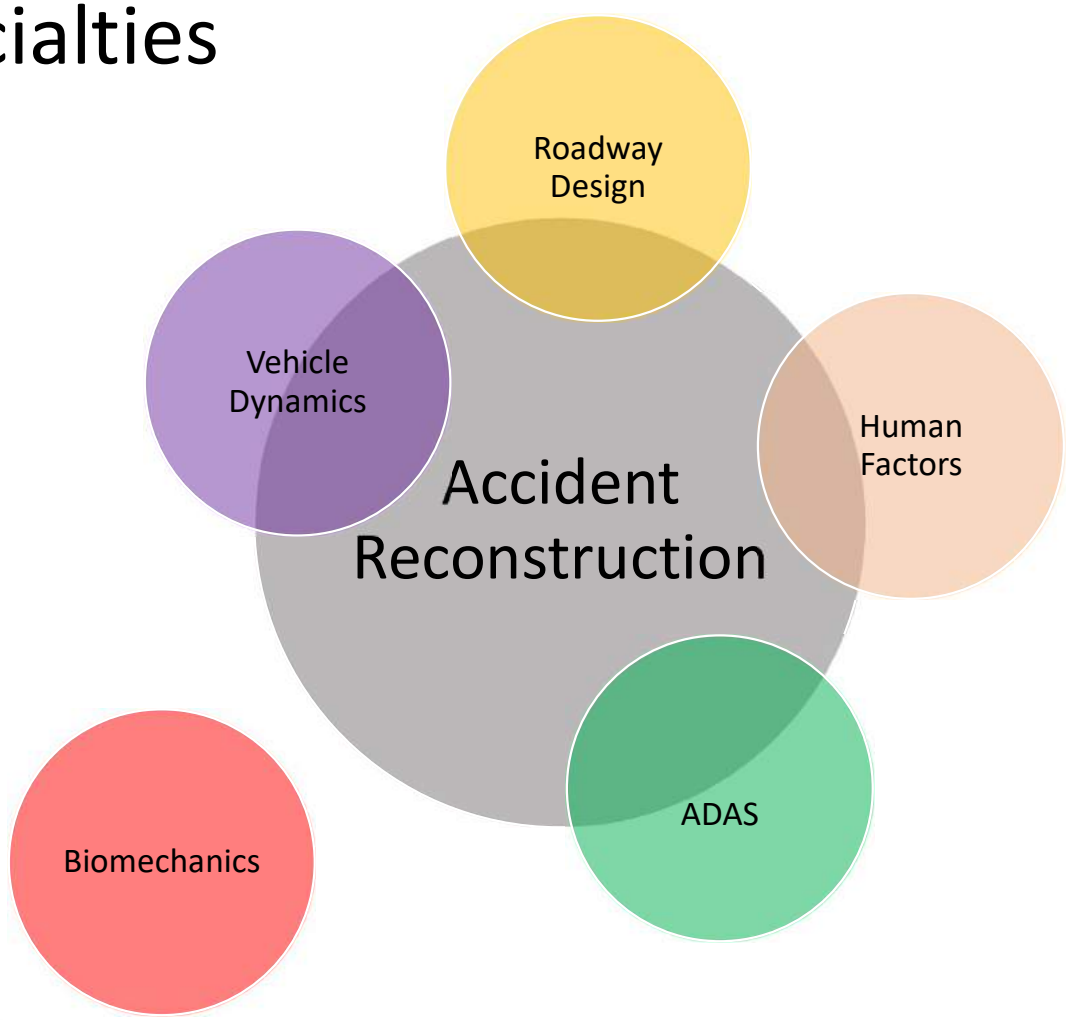
Figure 6: Predicted percentage of registered vehicles with front crash prevention



Predicted availability and fitment of safety features on registered vehicles, IIHS/HLDI, 2018

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Overlapping specialties



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Overview of the available technology

By the end of this module you will be able to:

- Describe what features comprise ADAS
- Describe the safety implications of the technologies

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Overview of the available technology

- Adaptive cruise control (ACC)
- Forward Collision Warning (FCW)
- Automatic Emergency Braking (AEB) with Brake Assist
- Blind spot monitoring and rear cross traffic alert
- Forward Cross-Traffic Alert
- Post-Impact Automatic Braking (PIB)

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Overview of the available technology

- Lane departure warning and lane keep assist (LDW, LKA)
- Lane centering
- Traffic signal awareness
- SAE Level 4 “Full Self Driving”

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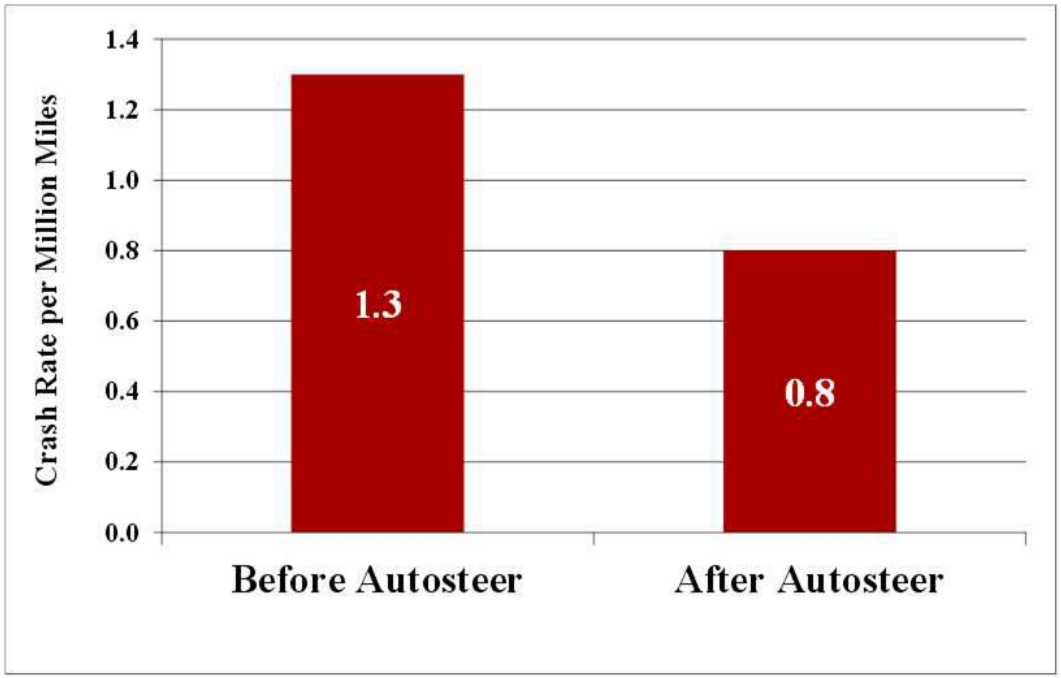
Effectiveness of FCW and AEB

- FCW alone, low-speed AEB, and FCW with AEB *reduced* rear-end striking crash involvement rates by 27%, 43%, and 50%, respectively[1]
- FCW with AEB *increased* rates of rear-end struck crash involvements by 20%[1]
- Model years 2010–2014

[1]Effectiveness of Forward Collision Warning and autonomous Emergency Braking Systems in reducing front-to-rear crash rates, Cicchino, Jessica B., Accident Analysis and Prevention February 2017

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Overview of the available technology



Tesla crash rates dropped 40% after introduction of Autopilot

[1]NHTSA Office of Defects Investigation #PE 16-007, 2017

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Overview of the available technology

Summary:

- ADAS is compromised of a variety of driver assistance systems
- ADAS provides safety improvements, but data is sparse due to recent adoption

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Levels of Self-Driving Vehicles

By the end of this module you will be able to:

- Describe the levels of autonomy
- Evaluate fallback performance

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SAE Levels of Automation

- Level 1: Cruise Control
- Levels 2-3: Adaptive cruise control and lane centering
- Levels 4-5: Hands-off operation
- The difference is the fallback performance; how much time is the driver given to respond?

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SAE level	Name	Narrative Definition	Execution of Steering and Acceleration/ Deceleration	*Monitoring* of Driving Environment	Fallback Performance of *Dynamic Driving Task*	System Capability *(Driving Modes)*
***Human driver* monitors the driving environment**						
0	**No Automation**	the full-time performance by the *human driver* of all aspects of the *dynamic driving task*, even when enhanced by warning or intervention systems	Human driver	Human driver	Human driver	n/a
1	**Driver Assistance**	the *driving mode*-specific execution by a driver assistance system of either steering or acceleration/deceleration using information about the driving environment and with the expectation that the *human driver* perform all remaining aspects of the *dynamic driving task*	Human driver and system	Human driver	Human driver	Some driving modes
2	**Partial Automation**	the *driving mode*-specific execution by one or more driver assistance systems of both steering and acceleration/ deceleration using information about the driving environment and with the expectation that the *human driver* perform all remaining aspects of the *dynamic driving task*	**System**	Human driver	Human driver	Some driving modes
***Automated driving system* ("system") monitors the driving environment**						
3	**Conditional Automation**	the *driving mode*-specific performance by an *automated driving system* of all aspects of the dynamic driving task with the expectation that the *human driver* will respond appropriately to a *request to intervene*	System	**System**	Human driver	Some driving modes
4	**High Automation**	the *driving mode*-specific performance by an automated driving system of all aspects of the *dynamic driving task*, even if a *human driver* does not respond appropriately to a *request to intervene*	System	System	**System**	Some driving modes
5	**Full Automation**	the full-time performance by an *automated driving system* of all aspects of the *dynamic driving task* under all roadway and environmental conditions that can be managed by a *human driver*	System	System	System	**All driving modes**

SAE J3016
Levels
of Automation

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Safety Elements of self-driving vehicles[1]



[1]NHTSA's voluntary guidance, Automated Driving Systems 2.0 – A Vision for Safety

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Levels of Self-Driving Vehicles

Summary:

- The driver must be vigilant if needed as the fallback
- Levels 2-3 require driver involvement
- Levels 4-5 do not require driver attention

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The enabling technologies

- By the end of this module you will be able to describe the technologies that enabled ADAS:
  - Sensors
  - Actuators
  - Computing

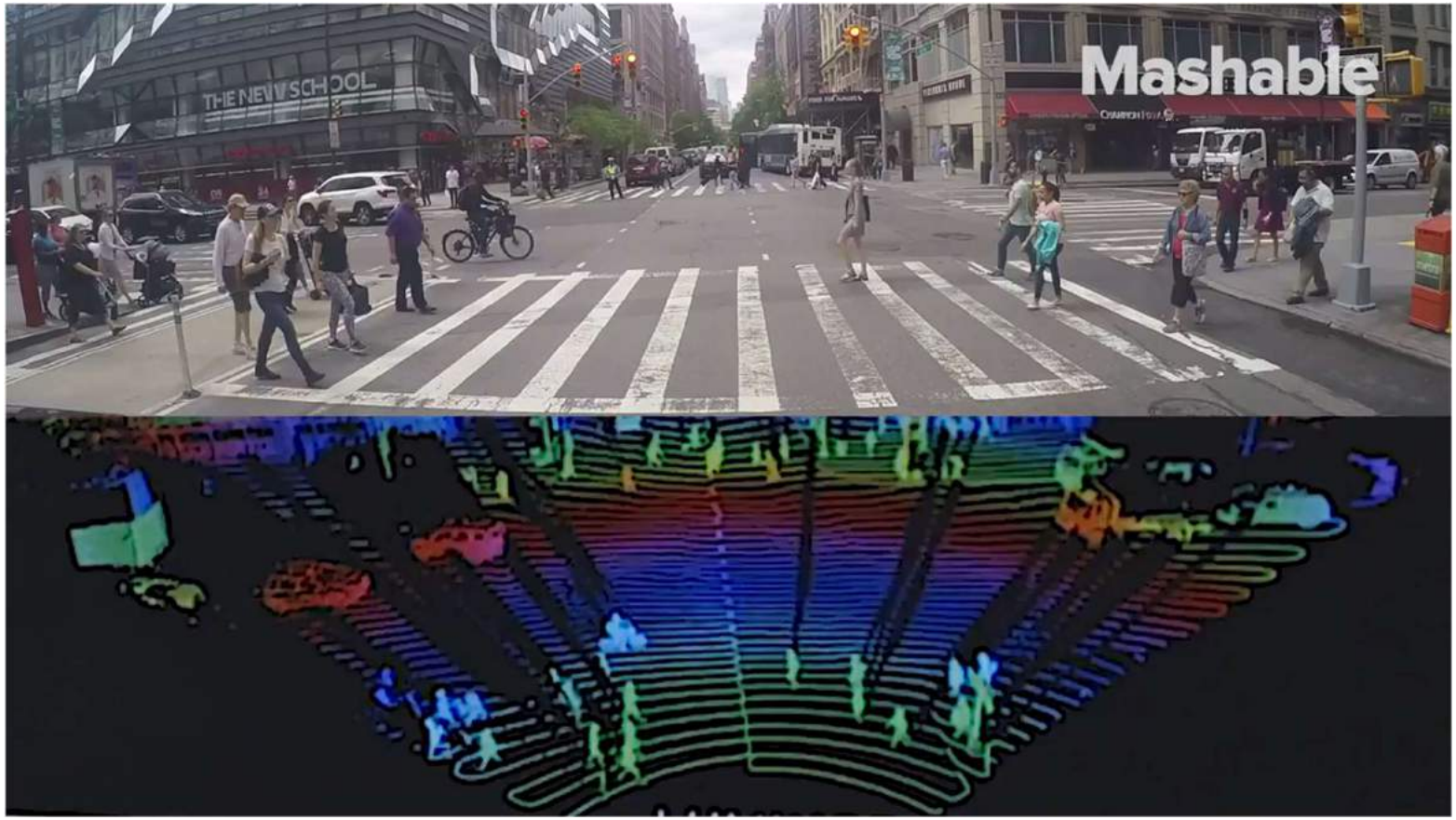
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Enabling Technologies: Sensors

- Cameras (including eye-tracking)
- GPS
- LIDAR*
- Radar*
- V2V/V2I*

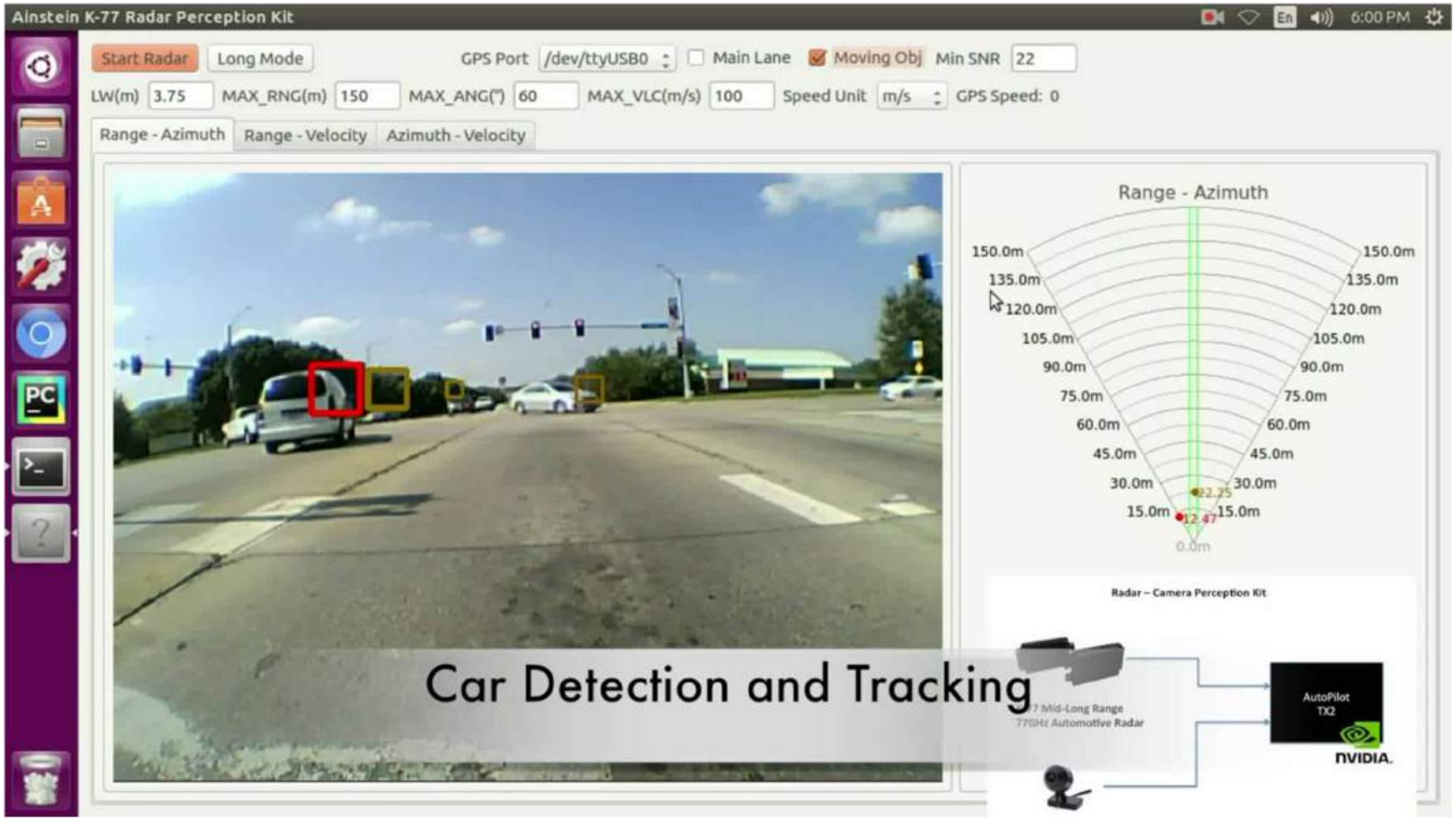
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LIDAR



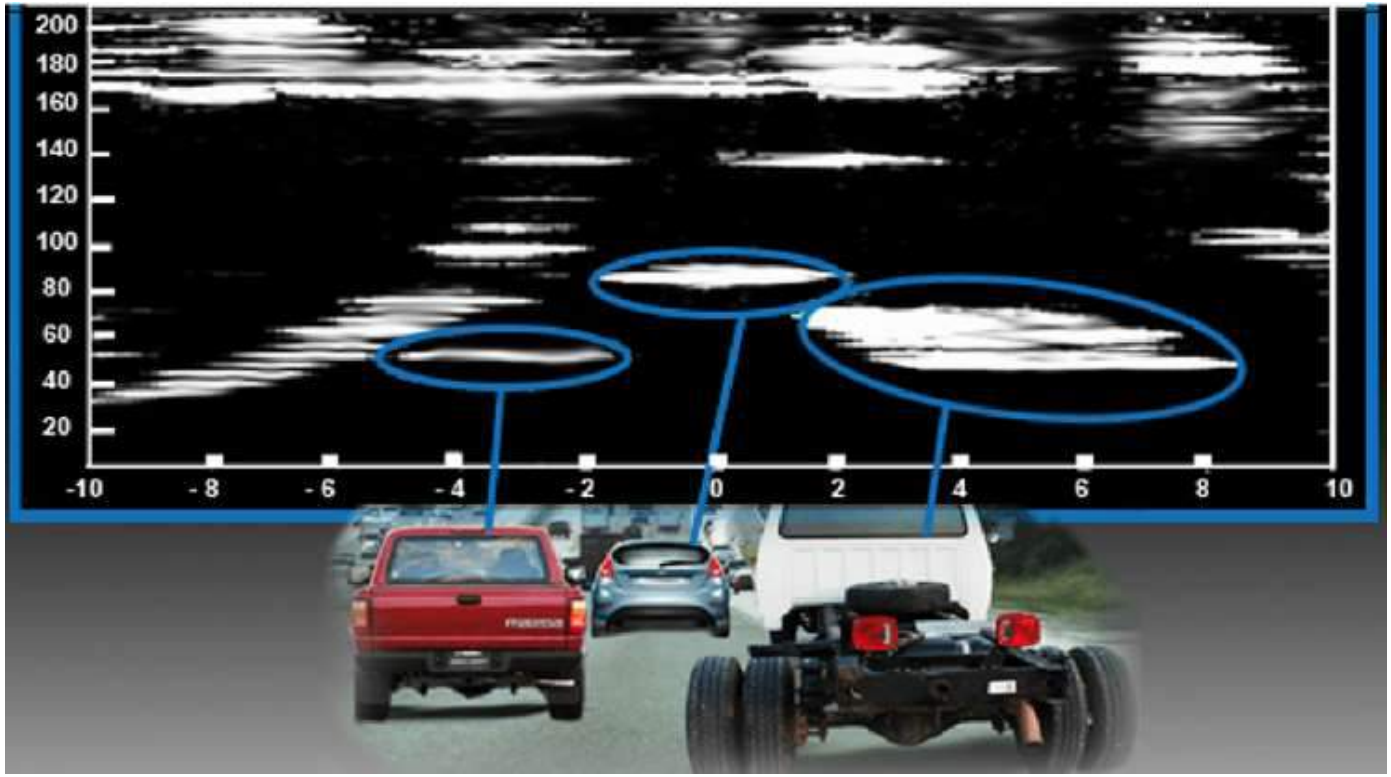
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Radar – Ainstein Video



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Radar Visualization



Collision Avoidance and Driver Support [34S34W1], Fordservicetraining.com, 2019

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V2I – Sirius XM Automatic Crash Notification (ACN+)

INCIDENT COMMENTS

DATE TIME:	USERID:	DEVICE:	COMMENTS:
[redacted]	BS11144	S05	Vehl 6|: Description: Vehicle
	BS15683	C16	Dup: SIRIUS XM ADV CRASH NOTIFICATION INVOLVING HONDA //NOI
	BS19781	C13	Dup: AIRBAGS DEPLOYED S/4.. UNK INJ CC TEXT: TRAFFIC COLLISION / TRANSPORTATION INCIDENT CAD RESPONSE: BRAVO DISPATCH LEVEL: 77B03 (UNKNOWN SITUATION/OTHER CODES NOT APPLICABLE) DISPATCH SUFFIX: T (TRAPPED/PINNED VICTIM) VEHICLE 1 INFO -- DESCRIPTION: VEHICLE -- COLOR: WHI -- MODEL: CARAVAN VEHICLE 2 INFO -- DESCRIPTION: VEHICLE -- COLOR: RED -- MODEL: MINI COOPER 1. AT LOC (2ND PTY). 2. TWO VEH. 4. VEH TYPE: WHI CARAVAN RED SPORTS CAR 3. 2 VEHS INVL. 5. NO VEHS ON FIRE. 6. ONE TRAPPED/PINNED. 7. UNK IF ANYONE THROWN OUT. 8. NOT ON HIGHWAY/FREEWAY/INTERSTATE. 9. NOT A DIFFICULT-TO-ACCESS AREA. 11. UNK IF ANY OBV INJ. 12. NO FUEL/FLUID LEAK. 13. OTHER HAZARDS INVL: DEBRIS 14. VEH BLOCKING TRAFFIC.

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Enabling Technologies: Actuators

- Electric Power Assist Steering (EPAS, EPS)
- Throttle by wire
- Electronic Stability Control (ESC) brake control

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Enabling Technologies: Computing

- Image classification – MobilEye and Bayes theorem
- Artificial intelligence and neural nets
- Probabilistic decision making

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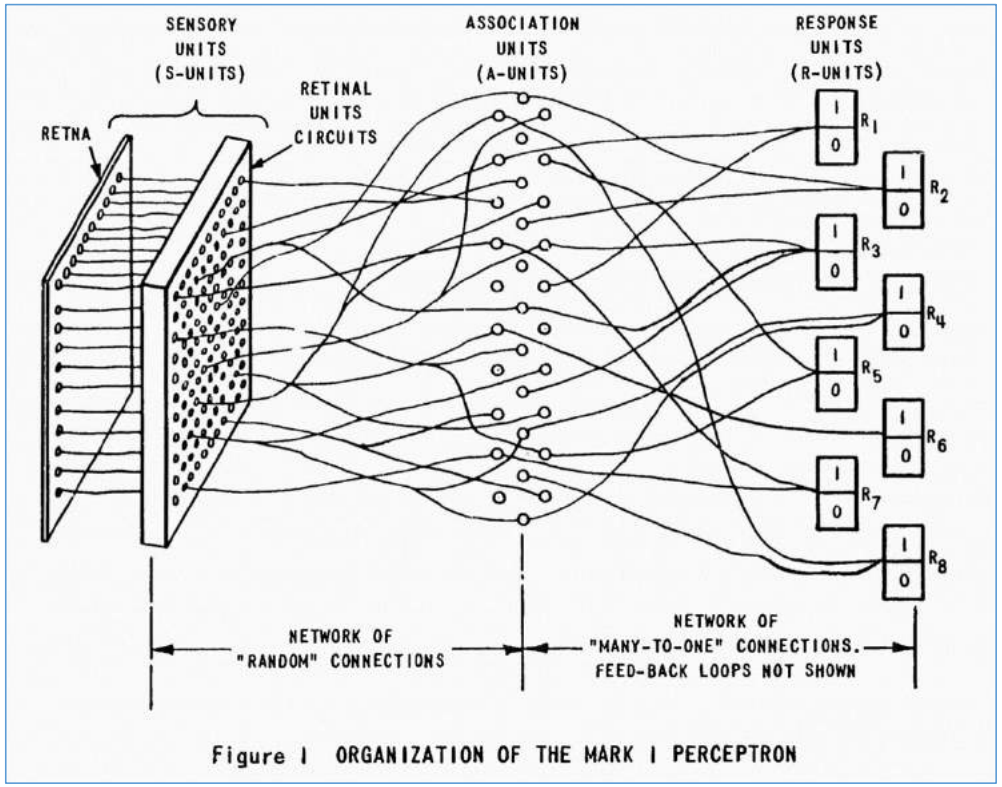
Image Classification



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1960s Image Classifier



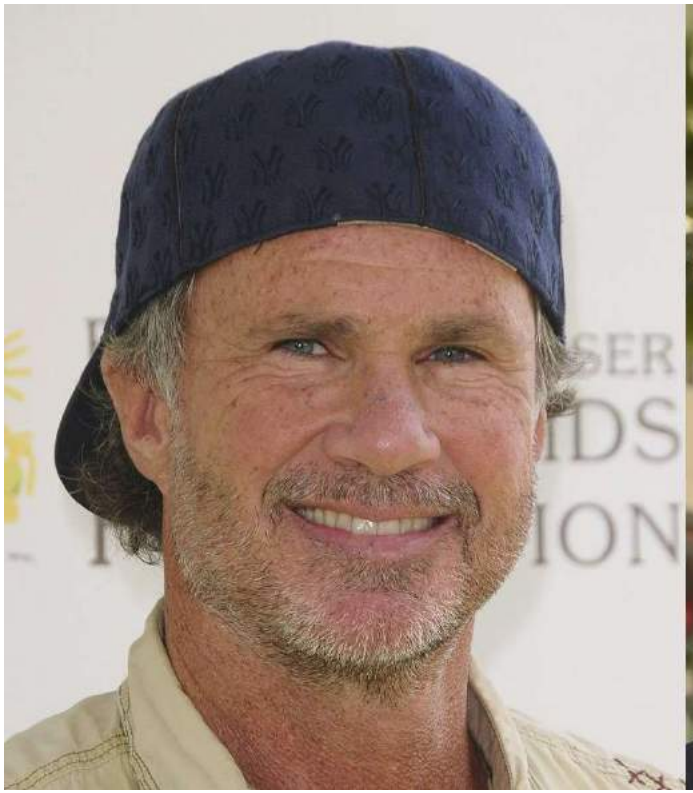


Mark 1 Perceptron Operator's Manual, Cornell Aeronautical Laboratory, 1960
https://apps.dtic.mil/dtic/tr/fulltext/u2/236965.pdf

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Image Classification and Artificial Intelligence

Chad Smith



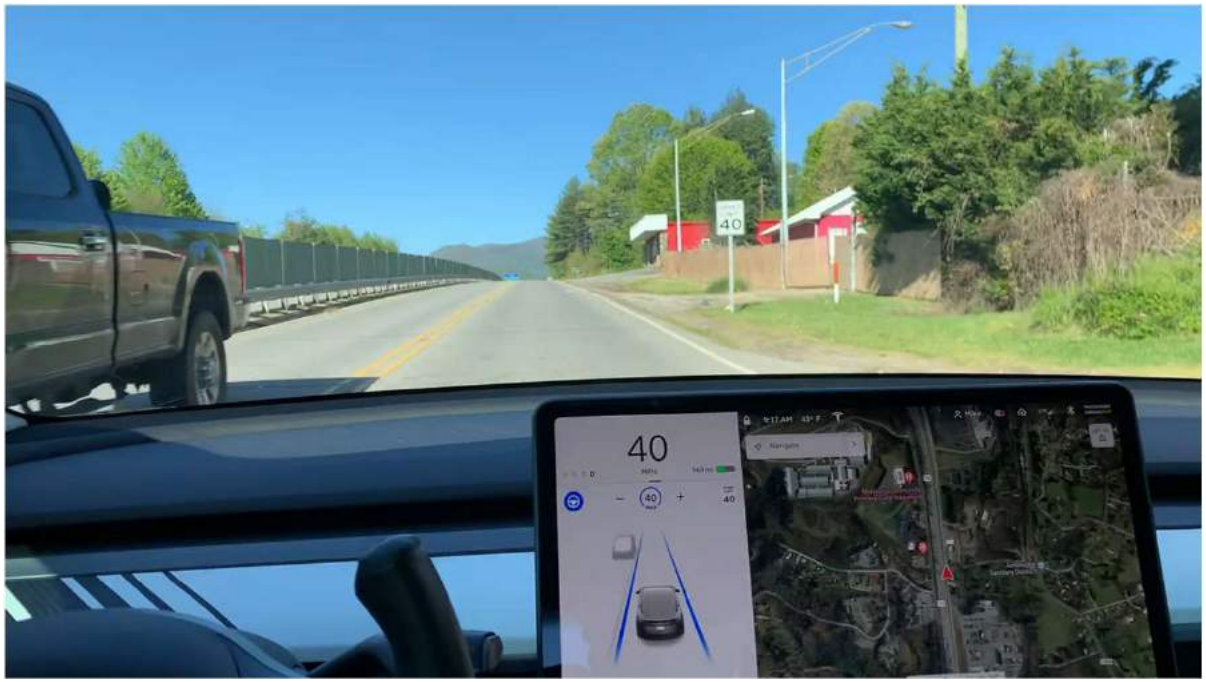


Will Farrell

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Will Ferrell (93%)
Chad Smith (56%)
Jimmy Fallon (83%)

Tesla Autopilot stops for Burger King



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Tesla Autopilot Visualization



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Tesla “Full Self Driving” Visualization



Tesla Neural Network Recruiting Demo
https://www.tesla.com/autopilotAI

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The enabling technologies

- Summary:
- Technology in three areas – sensors, actuators, and computing have allowed ADAS to enter the market

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What is on the road today?

By the end of this module you will:

Know what technology is in current production vehicles, and what some of the limitations are.

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Subaru “Eyesight” stereo cameras





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Cadillac SuperCruise

- Adaptive cruise and lane centering
- “Selfie” camera monitors driver attention



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Tesla

- Adaptive Cruise and lane centering
- Lane change assist
- Requires steering wheel feedback at irregular intervals
- Audible and visual alarm at fallback (sometimes)



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Toyota

90% of U.S. models 2018+ equipped with Forward Collision Warning

Records copious amounts of data available through Vehicle Control History (VCH) and Freeze Frame Data (FFD)

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Ford – Evasive Steering Assist



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What is on the road today?

Summary:

OEMs have implemented ADAS technology in different manners, all with the same goal: keep the driver involved while providing assistance in the driving task.

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Standards & Protocols

By the end of this module you will:

- Know what relevant standards have been promulgated by SAE, ISO, NHTSA, IIHS, and Euro NCAP
- Have perspective on the rapid changes occurring in the field

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IIHS and Euro NCAP AEB Target

The ADAC Advanced Emergency Braking System (AEBS), which is produced by Messring Systembau GmbH, is used as the stationary target (http://www.messring.de/test-facilities-and-components/adac/). The stationary carriage and balloon car are shown in Figure 2. The balloon car is equipped with a U.S.-specific license plate. At the beginning of testing, the target will be inflated to 250 mbar and maintained at that pressure throughout testing. The target cover should also be free of wrinkles during testing.

Figure 2

ADAC Advanced Emergency Braking System Stationary Target



IIHS Autonomous Emergency Braking Test Protocol (Version I), 2013

Euro NCAP Test Protocol – AEB Systems, Version 1.1, 2015

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Deformable Barrier – Soft Car 360





https://www.youtube.com/watch?v=BFfqzfTqvis

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Accident Reconstruction Test Barriers





MEA Forensic Engineers & Scientists

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ISO 22839 - Forward vehicle collision mitigation systems

6.3.6.4.1.1 Light vehicles

MB shall not be initiated for TTC or ETTC above 3,0 s to achieve the deceleration requirement of 6.3.6.4.2.

6.3.6.4.2.1 Light vehicles

FVCMS shall generate a deceleration at least 5,0 m/s^2 (0,51 g) for a duration chosen to achieve a speed reduction of at least 2,0 m/s. FVCMS with both MB and SRB combined (Type 3) shall generate a minimum speed reduction of at least 4,0 m/s. This requirement does not constrain the time when mitigation braking is activated.

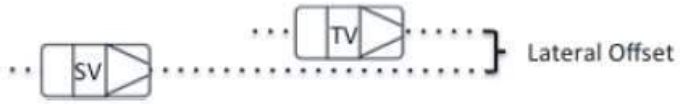
0.51G

9 mph

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ISO 22839 – Automatic Emergency Braking

3.20
lateral offset
lateral distance between the longitudinal centerlines of a subject vehicle (SV) and a target vehicle (TV), measured as a percentage of the width of the SV, such that if the centers of the two vehicles are aligned, the value is zero



6.3.3.2.3 Maximum lateral offset (lateral discrimination)

FVCMS shall function when the lateral offset is 20 % or lower in either direction. FVCMS may function for lateral offsets greater than 20 % at the discretion of the manufacturer.

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NHTSA AEB Procedure (draft, 2014)

Table 2-2 CIB and DBS Draft Assessment Reference Values (ARVs)

AEB System	LVS 25_0	LVM 45_20	LVM 25_10	LVD1 35_35	LVD2 25_25	STP_45 (FP)	STP_25 (FP)
CIB	Speed Reduction ≥ 9.8 mph (15.8 km/h)	Speed Reduction ≥ 9.8 mph (15.8 km/h)	Crash Avoidance	Speed Reduction ≥ 10.5 mph (16.9 km/h)	Speed Reduction ≥ 9.8 mph (15.8 km/h)	No Activation[1]	
DBS	Crash Avoidance					No Activation[2]	

[1]STP = Steel trench plate

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Autobrake Activation on Steel Trench Plate



AEB False positive, captured by Comma.AI: https://comma-ai.medium.com/aeb-a-case-study-using-comma-ai-dataset-2fc08a2397f4

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IIHS AEB Rating Scale

	12 mph test			25 mph test				Forward collision warning
Speed reduction (mph)	less than 5	5 to 9	10 or more	less than 5	5 to 9	10 to 21	22 or more	n/a
Points	0	1	2	0	1	2	3	1

IIHS Frontal Crash Prevention Ratings, 2016

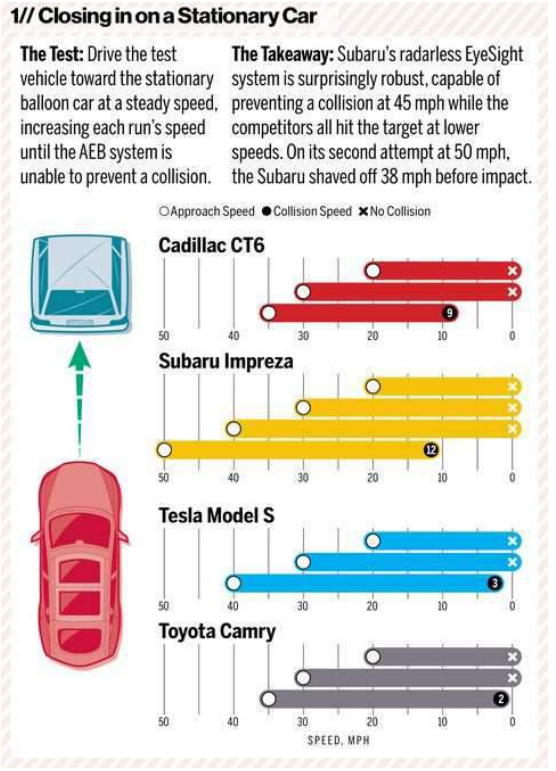
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Car & Driver AEB Testing, 2018

- Tested vehicles can avoid a stopped vehicle when traveling 20-30mph, and up to 40mph in some cases.

- Maximum braking of tested vehicles:
  - .4-.7g on Adaptive Cruise
  - .9-1g Autobrake

- Inconsistent performance



We Crash Four Cars Repeatedly to Test the Latest Automatic Braking Safety Systems, Tingwall, 2018
https://www.caranddriver.com/features/a24511826/safety-features-automatic-braking-system-tested-explained/

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Standards & Protocols

Summary:

- Standards have been promulgated by SAE, ISO, NHTSA, IIHS, and Euro NCAP
- The standards provide a preliminary framework in a rapidly changing field

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Human Factors

By the end of this module you will:

- Understand that human-machine interaction is key to public trust in autonomy
- Gain further knowledge of necessary driver involvement in Level 2-3 vehicles

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Human-Machine-Interface

Augmented Reality will be Key for Driver Acceptance



From Assisted to Automated Driving, Alfred Eckert, Continental Chassis & Safety Division, 2015

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HMI - Tesla



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HMI – Detroit Diesel



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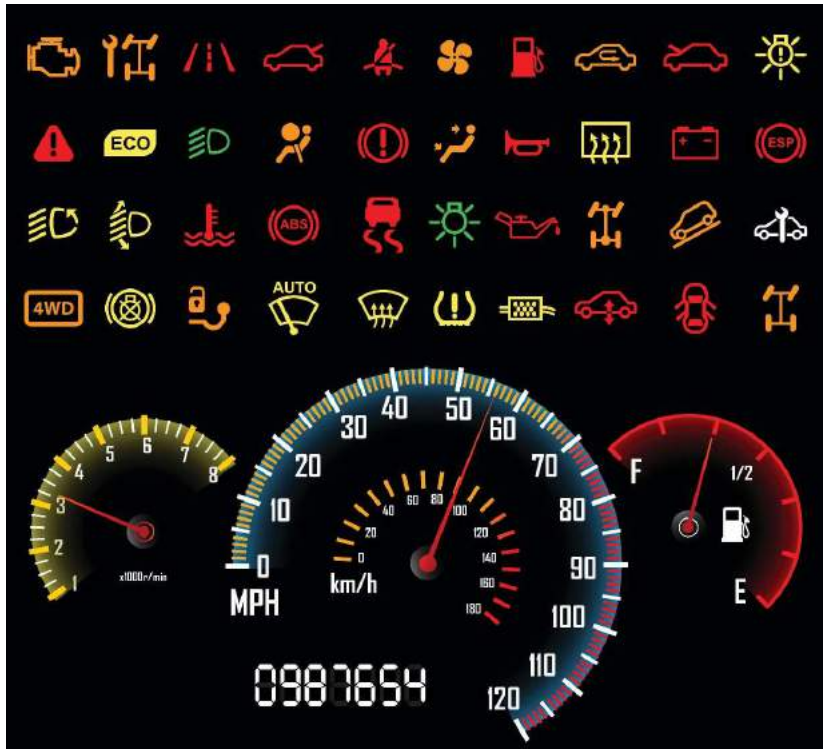
Human factors – sharing the road



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Driver braking in response to audible/visual alert

0.8s from known sound to known response (foot on the brake pedal).



IDRR Software and Related Research, Jeff Muttart and Swaroop Dinakar, 2019

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Mis-use of automation – SAE J3114

Abuse: The intentional mis-use of a driving automation system

Eutactic Behavior – Use outside of design intent

Disuse – usually the result of a complete lack of trust

Over-reliance, bias & complacency

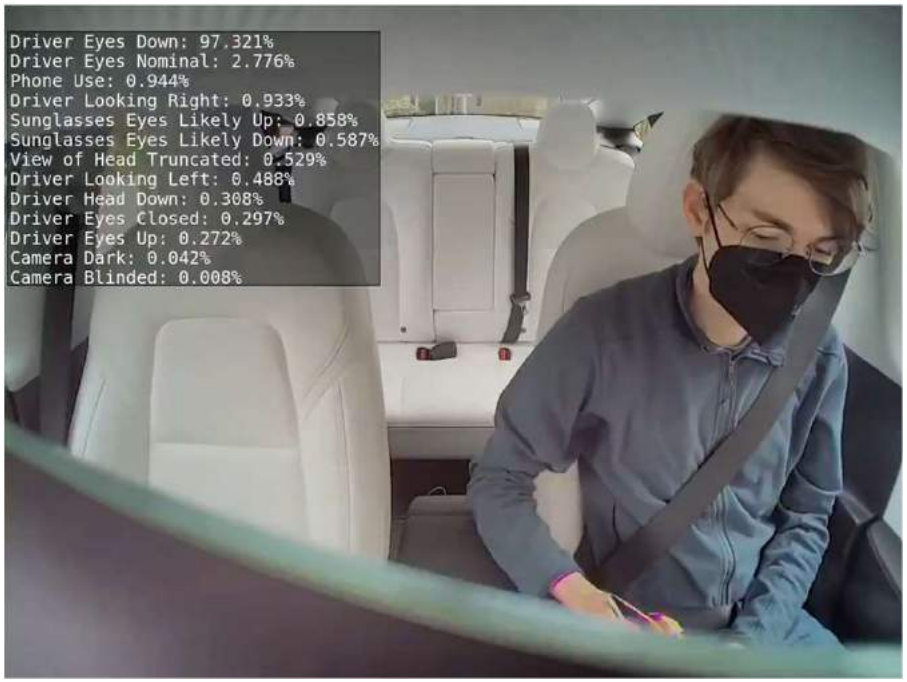
Lack of receptivity – failure to respond to automation warning





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Tesla Model 3/Y Driver Monitoring System



https://twitter.com/greentheonly/status/1379928419136339969?s=20

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Human Factors

Summary:

- Human-machine interaction is key to public trust in autonomy
- Driver involvement is necessary in Level 2-3 vehicles
- Drivers need to time to re-engage with the vehicle and environment after a period of autonomous operation

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Liability and Litigation

By the end of this module you will:

- Know the current and future status of ADAS reconstruction
- Have a method for evaluating ADAS-involved accidents

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Liability and litigation – current & future status

- Current status: Drivers can blame technology for what may be driver error
  - "To a user who experiences it, a driving automation system performance limitation may be indistinguishable from a driving automation system failure"[1]

- Manufacturers can blame the driver for abuse and eutactic behavior (not complying with design intent and/or owner's manual)

- Future status: Manufacturers may accept liability if vehicle was operating within design parameters. Insurance rates may lower.

1 Human Factors Definitions, SAE J3114, 2016

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Forensic Approach to ADAS Accidents

Investigation: Was the vehicle equipped with ADAS technology?

Analysis: Did it perform as designed?

Effectiveness: Could it have mitigated or prevented the accident?

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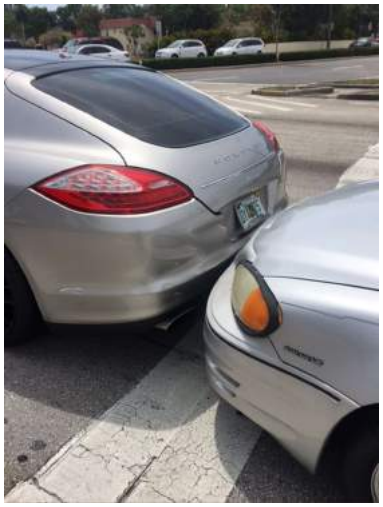
2019 Ford Transit – Camera opening with plastic insert





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Liability and litigation – forensic approaches






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Liability and litigation – forensic approaches



Driver Awareness Package

- Safety Alert Seat
- Forward Collision Alert
- Low Speed Forward Automatic Braking
- Lane Keep Assist with Lane Departure Warning
- Lane Change Alert
- Rear Cross Traffic Alert
- Side Blind Zone Alert[14]
- IntelliBeam automatic high-beam On/Off

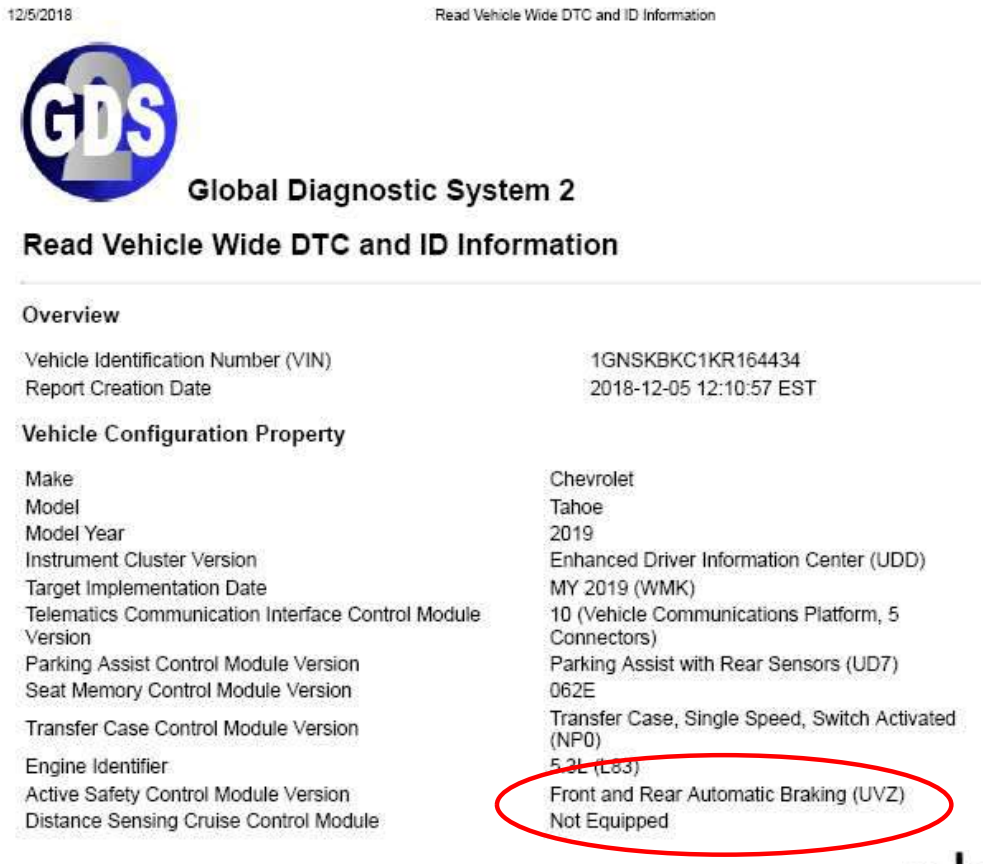
If the vehicle has Forward Collision Alert (FCA), it also has FAB, which includes Intelligent Brake Assist (IBA).

FAB and IBA can be disabled through vehicle personalization. See "Collision/

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Build sheets and build data

- Compnine.com
- GDS2 (For GM Vehicles)



12/5/2018 Read Vehicle Wide DTC and ID Information

Global Diagnostic System 2

Read Vehicle Wide DTC and ID Information

Overview

Vehicle Identification Number (VIN)	1GNSKBKC1KR164434
Report Creation Date	2018-12-05 12:10:57 EST

Vehicle Configuration Property

Make	Chevrolet
Model	Tahoe
Model Year	2019
Instrument Cluster Version	Enhanced Driver Information Center (UDD)
Target Implementation Date	MY 2019 (WMK)
Telematics Communication Interface Control Module Version	10 (Vehicle Communications Platform, 5 Connectors)
Parking Assist Control Module Version	Parking Assist with Rear Sensors (UD7)
Seat Memory Control Module Version	062E
Transfer Case Control Module Version	Transfer Case, Single Speed, Switch Activated (NP0)
Engine Identifier	5.3L (L83)
Active Safety Control Module Version	Front and Rear Automatic Braking (UVZ)
Distance Sensing Cruise Control Module	Not Equipped

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Liability and litigation – forensic approaches

Pre-Crash Data -5.0 to -0.5 sec (Event Record 1)

Times (sec)	Accelerator Pedal, % Full (Accelerator Pedal Position)	Service Brake (Brake Switch Circuit State)	Engine RPM (Engine Speed)	Engine Throttle, % Full (Throttle Position)	Speed, Vehicle Indicated (Vehicle Speed) (MPH [km/h])
-5.0	0	Off	1408	8	35 [57]
-4.5	0	Off	1344	8	35 [56]
-4.0	0	On	1280	11	35 [56]
-3.5	0	On	1152	8	34 [55]
-3.0	0	On	1152	8	34 [54]
-2.5	0	On	1152	8	33 [53]
-2.0	0	On	1088	8	32 [52]
-1.5	0	Off	1088	8	32 [51]
-1.0	0	On	1024	8	31 [50]
-0.5	0	On	832	8	24 [39]

0.64G, 7mph dV

2017 Cadillac Escalade ESV Platinum

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ADAS Simulation Options

- PreScan with MatLab/Simulink
- CARLA
- LGSVL
- Gazebo
- CarSim

"Our analysis shows that CARLA and LGSVL simulators are the current state-of-the art simulators for end to end testing of self-driving cars for the reasons mentioned in this paper"

A Survey on Simulators for Testing Self-Driving Cars – P. Kaur, et al 2021.
https://arxiv.org/pdf/2101.05337.pdf

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Liability and Litigation

- Summary:
- Currently, drivers may attempt to blame ADAS features for causing or not preventing an accident
- You have a method for evaluating ADAS-involved accidents

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Electronic Data (EDR)

By the end of this module, you will:

- Know what EDR is available, and how it can be used in ADAS-involved accidents
- Have a workflow for manufacturer-specific EDRs

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2018 Ford Edge Data Elements

Pre-Crash Data -5 to 0 sec [5 samples/sec] (First Record) - Table 1 of 2

Time (sec)	Brake Pre-Charge Request	Brake Assist Sensitivity Level	Brake Deceleration Request (m/s^2)	Brake Deceleration Request Enable	Large Driver Steering or Accel Pedal Input	Collision Mitigation System Fault
- 5.0	No PreCharge Request	Normal	-20	No	No	No
- 4.8	No PreCharge Request	Normal	-20	No	No	No
- 4.6	No PreCharge Request	Normal	-20	No	No	No
- 4.4	No PreCharge Request	Normal	-20	No	No	No
- 4.2	No PreCharge Request	Normal	-20	No	No	No
- 4.0	No PreCharge Request	Normal	-20	No	No	No

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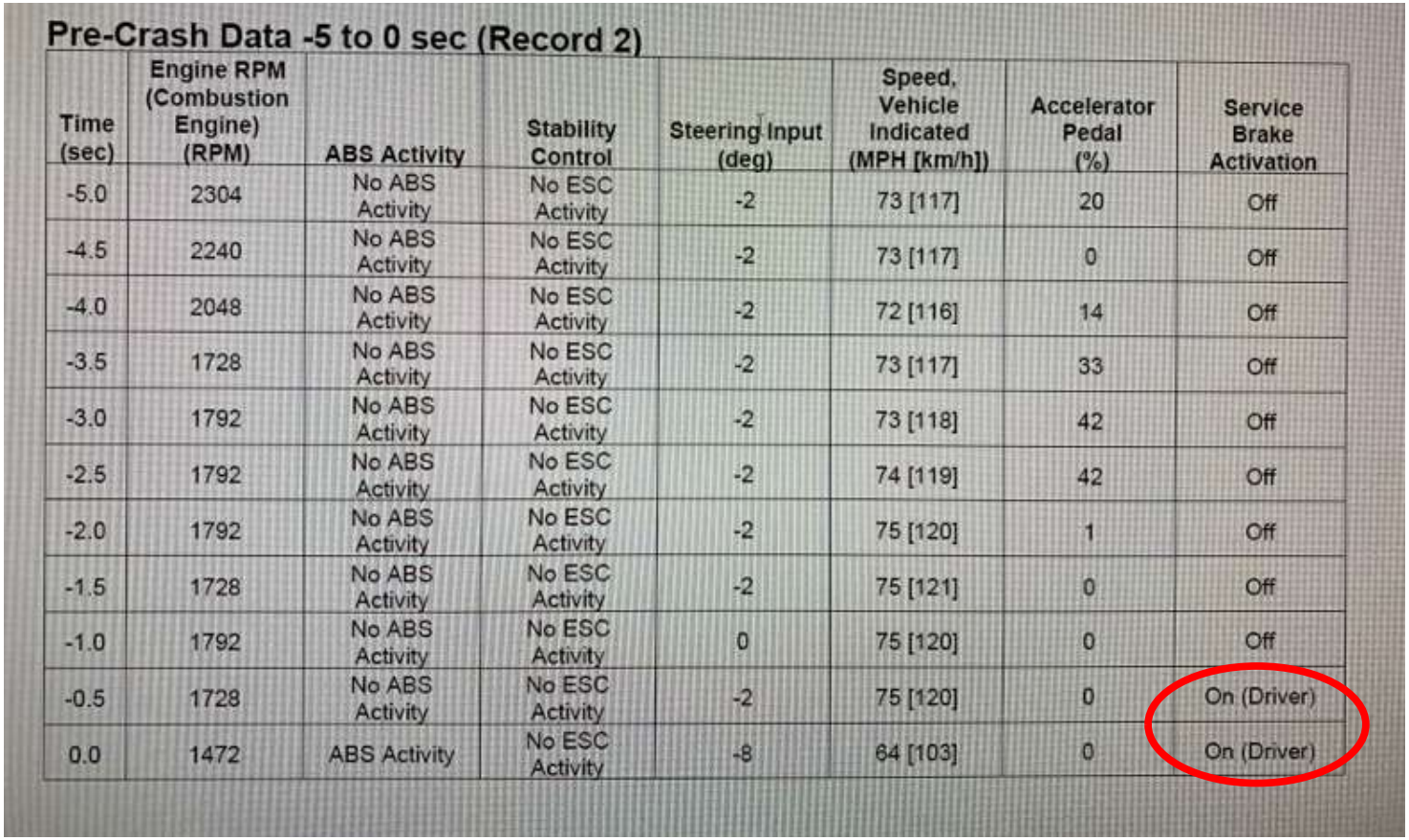
2018 Chrysler 300 CDR Data Limitations

Master Cylinder Pressure - This indicates the brake pressure applied to the brakes through the brake pedal.

Service Brake - "On" indicates that the brake pedal is physically depressed. Braking from the ABS or FCW systems will not be reported in this data element.

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2018 Audi Data Elements



Pre-Crash Data -5 to 0 sec (Record 2)

Time (sec)	Engine RPM (Combustion Engine) (RPM)	ABS Activity	Stability Control	Steering Input (deg)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal (%)	Service Brake Activation
-5.0	2304	No ABS Activity	No ESC Activity	-2	73 [117]	20	Off
-4.5	2240	No ABS Activity	No ESC Activity	-2	73 [117]	0	Off
-4.0	2048	No ABS Activity	No ESC Activity	-2	72 [116]	14	Off
-3.5	1728	No ABS Activity	No ESC Activity	-2	73 [117]	33	Off
-3.0	1792	No ABS Activity	No ESC Activity	-2	73 [118]	42	Off
-2.5	1792	No ABS Activity	No ESC Activity	-2	74 [119]	42	Off
-2.0	1792	No ABS Activity	No ESC Activity	-2	75 [120]	1	Off
-1.5	1728	No ABS Activity	No ESC Activity	-2	75 [121]	0	Off
-1.0	1792	No ABS Activity	No ESC Activity	0	75 [120]	0	Off
-0.5	1728	No ABS Activity	No ESC Activity	-2	75 [120]	0	On (Driver)
0.0	1472	ABS Activity	No ESC Activity	-8	64 [103]	0	On (Driver)

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2015 BMW X5 EDR

- CDR Data Limitations: “Service Brake Status only indicates driver initiated braking. An automatic braking (e.g. brake intervention by Adaptive Cruise Control) will not be recorded.”

- NHTSA NASS Case #N-2015-04-092-V2-ACM

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GM Front Camera Module (FCM)

Event Data (Record 1 - Pedestrian Warning)

Data Element	Pre-Event	At-Event	Post-Event
FCM Time (sec)	175.00	179.65	183.65
Vehicle Speed Average Non Driven (MPH [km/h])	41 [65]	41 [66]	43 [69]
Vehicle Acceleration - Longitudinal (m/s^2)	-0.07	-0.05	0.14
Vehicle Acceleration - Lateral (m/s^2)	0.50	1.00	0.25
Yaw Rate (deg/sec)	0.1250	0.4375	0.0625
VSES Active	False	False	False
ABS Active	False	False	False
FCA Alert Indication	None	None	None
LDW Alert Direction	No Alert	No Alert	No Alert
Accelerator Pedal, % Full (%)	17	18	18
Brake Pedal Position (%)	2	2	2
Service Brake (Brake Switch Circuit State)	False	False	False
Turn Signal Switch Active	No Activation	No Activation	No Activation
Adaptive Cruise Control System Selected Mode	No Cruise Control Active	No Cruise Control Active	No Cruise Control Active
AEB Control Automatic Braking Request Type	None	None	None
AEB Object Type	Car	Ped Child	Car
AEB Object Dynamic Property	Target Never Identified as Moved	Target Never Identified as Moved	Target Never Identified as Moved
AEB Object Longitudinal Distance (m)	0.0	14.1	0.0
AEB Object Relative Longitudinal Velocity (m/s)	0.0075	0.0075	0.0075
AEB Object Relative Longitudinal Acceleration (m/s^2)	0.000	0.000	0.000
AEB Object Lateral Distance (m)	0.0	0.0	0.0
AEB Object Azimuth (deg)	0	0	0
GPS Heading (deg)	88.3	88.7	88.3



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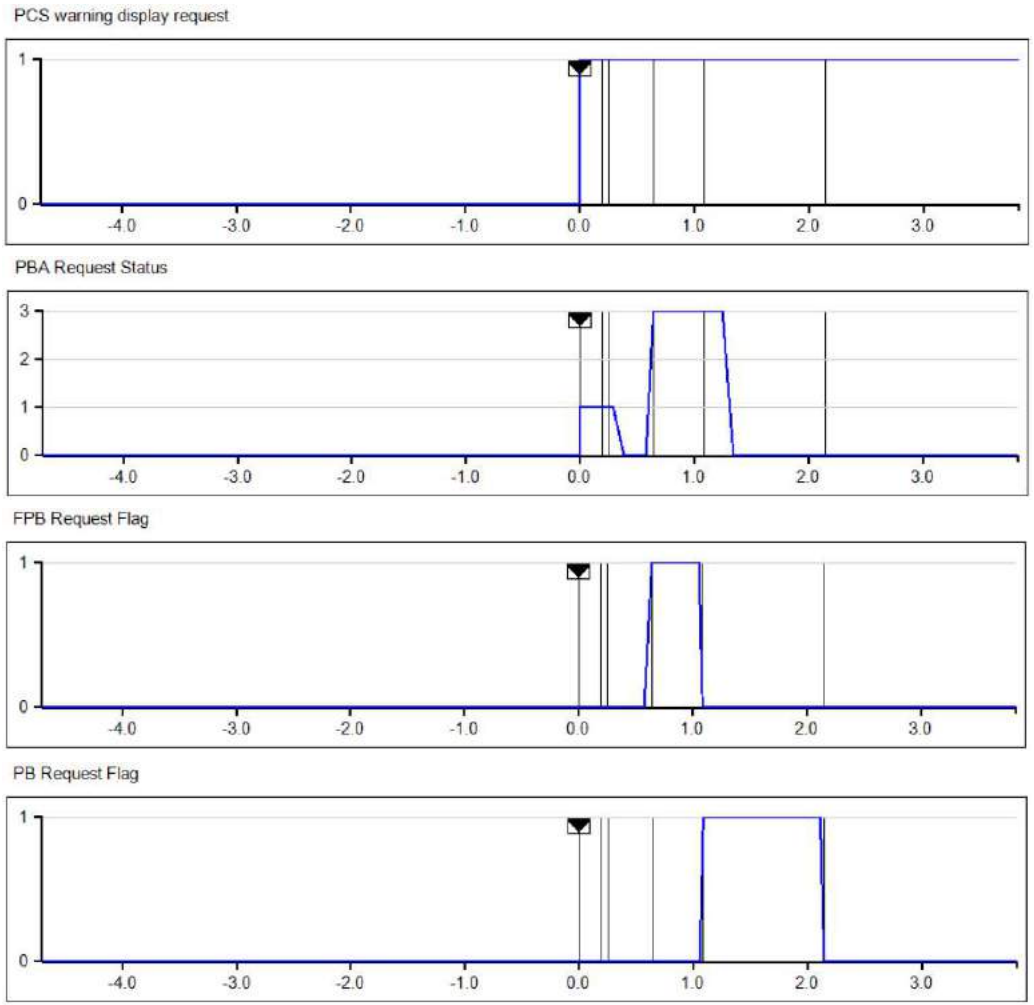
AEB Test Results – 2018 Toyota RAV4



Parameter	Value	Unit
Vehicle Speed	21	MPH
Brake SW	OFF	
Longitudinal G Sensor Signal	-0.28712	m/s2
ALM Request Flag	OFF	
PBA Request Flag	OFF	
FPB Request Flag	OFF	
PB Request Flag	OFF	
Target Distance	133.60	feet

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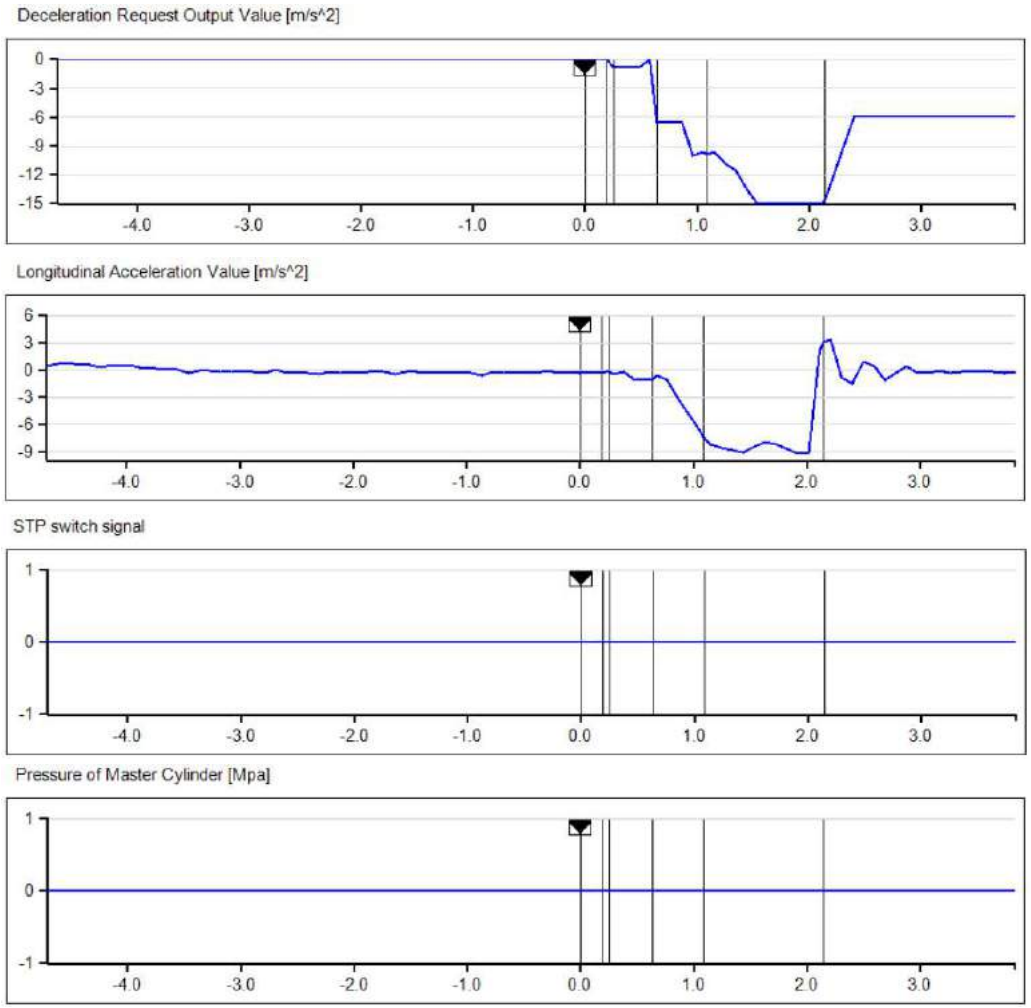
AEB Test Results – 2018 Toyota RAV4



Warnings and AEB status

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AEB Test Results – 2018 Toyota RAV4



Vehicle deceleration

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Toyota Image Sequence Example



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Green Light in VCH Image



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Bendix Wingman – False Positive

Time	Vehicle Speed (mph)	Steering Angle (deg)	Accelerator Pedal Position (%)	Brake Light	Driver Brake Application	Radar Collision Warning	ABS Activity
0	56	5	0	2	1	1	0
0.5	56	11	0	2	1	1	0
1	54	5	0	2	1	0	0
1.5	54	0	0	2	1	0	0
2	51	-11	0	2	1	0	0
2.5	51	-11	0	2	0	0	0
3	51	-16	38	0	0	0	0

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Bendix Wingman – False Positive



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Detroit Assurance 2 - AEB

	Object [the Toyota] Detected	Optic Accoustic Warning [Light and Alarm]	Haptic Braking [Brake pulsation]	Emergency Braking [Performed by computer]	Object [the Toyota] lost
Time (in Seconds)	-2.3	0	0.96	2.36	4.49
My Speed (in mph)	69.46	69.47	69.47	69.49	66.5
My acceleration (m/s2)	0	0	0	0	-2.5
My Steering angle (in degrees)	N/A	-5.32	-1.06	4.76	N/A
ABS Activity	0	0	0	0	0
ESP Activity	0	0	0	0	1
Tracked Object Speed (in mph)	25.13	15	10.73	4.11	2
Tracked Object Acceleration (in m/s2)	-2.2	-1.9	-1.9	-2.1	0
Distance to tracked object (in ft)	614.88	443.25	362.35	231.53	18.28
Lateral offset of tracked object (in ft)	0	-3.94	-2.62	-1.31	-7.22
ABA Availability (Y/N)	1	1	1	1	1
Driver Activity Occured	N/A	Yes	Yes	Yes	Yes

Wabco OnGuard (Radar Only)



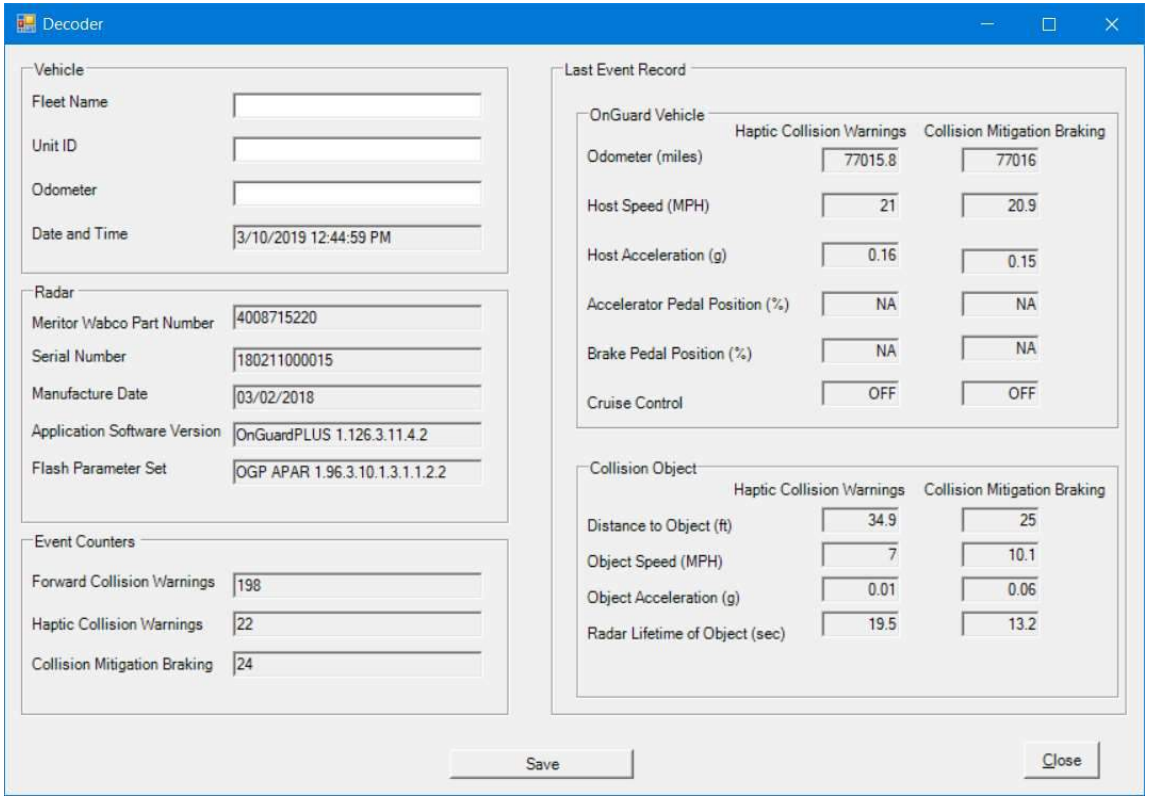
21mph approach to slow object

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WabCo OnGuard (Radar Only)



In WabCo Toolbox, click on OnGuard, then Parameter Decoder



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Wabco OnGuard (Radar Only)

OnGuard Event Log Fleet Report

Vehicle:-
Fleet Name: Ryder 2
Unit ID: 030919 1444
Odometer: Dash odo 77016
Date & Time: 3/9/2019 2:45:02 PM

Radar:-
Meritor Wabco Part Number: 4008715220
Serial Number: 180211000015
Manufcture Date: 03/02/2018
Application Software Version: OnGuardPLUS 1.126.3.11.4.2
Flash Parameter Set: OGP APAR 1.96.3.10.1.3.1.1.2.2

Event Counters:-
Forward Collision Warnings: 198
Haptic Collision Warnings: 22
Collision Mitigation Braking: 24

Event Records:-

{OnGuard Vhehicle}	[Haptic Warning]	[Collision Mitigation]
Odometer(miles):	77015.8	77016
Host Speed(MPH):	21	20.9
Host Acceleration(g):	0.16	0.15
Accelerator Pedal Position(%):	NA	NA
Brake Pedal Position(%):	NA	NA
Cruise Control:	OFF	OFF

{Collision Object}	[Haptic Warning]	[Collision Mitigation]
Distance to Object(ft):	34.9	25
Object Speed(MPH):	7	10.1
Object Acceleration(g):	0.01	0.06
Radar Lifetime of Object(sec):	19.5	13.2

Report saved from OnGuard Parameter Decoder

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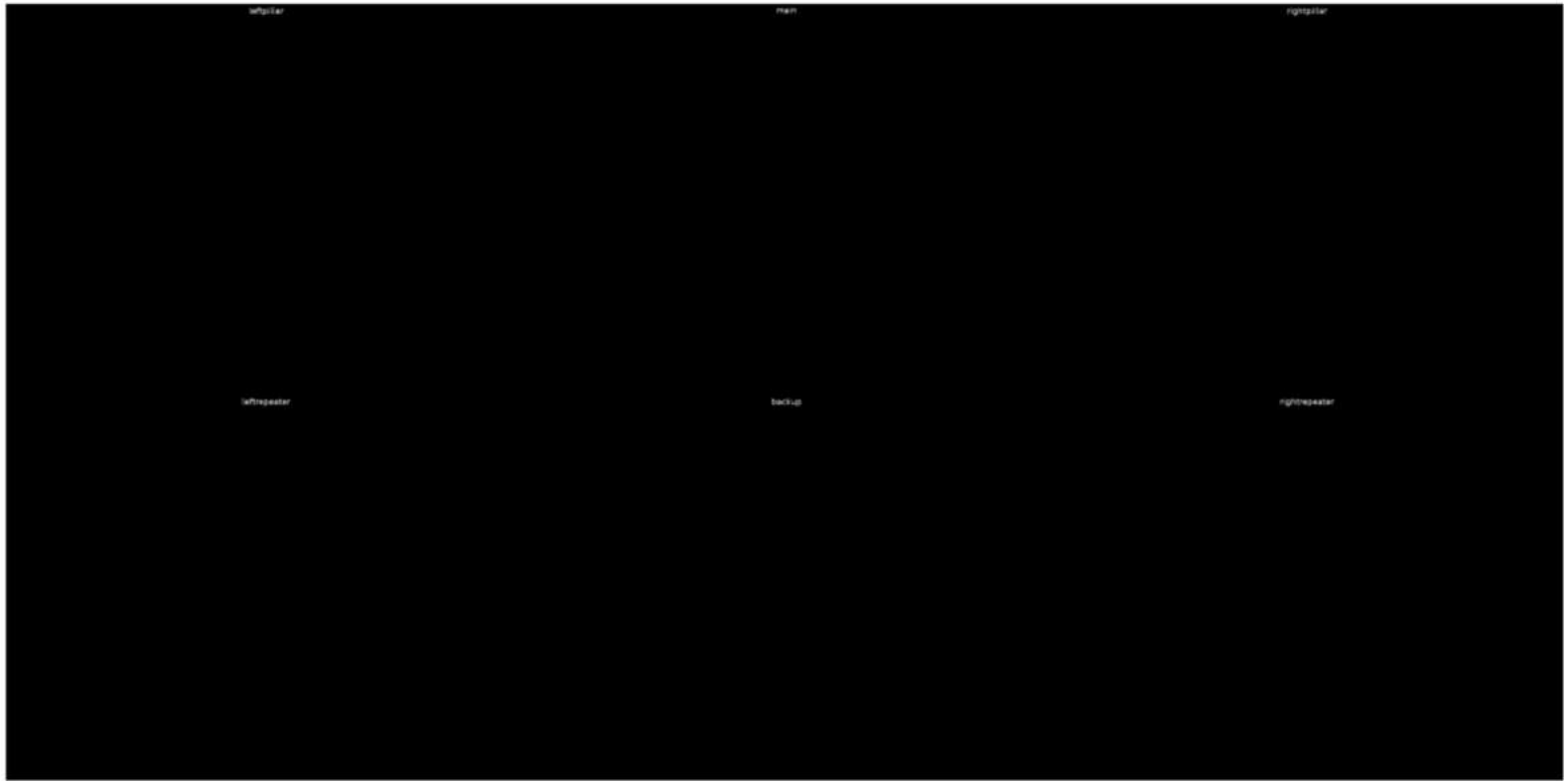
Tesla Model 3 DashCam Sample



https://www.youtube.com/watch?v=pa1U2mnTgBk

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Tesla Model 3 log video, extracted from SD Card



https://github.com/NetherlandsForensicInstitute/teslalogs/blob/master/data/snapshot_interaction.gif

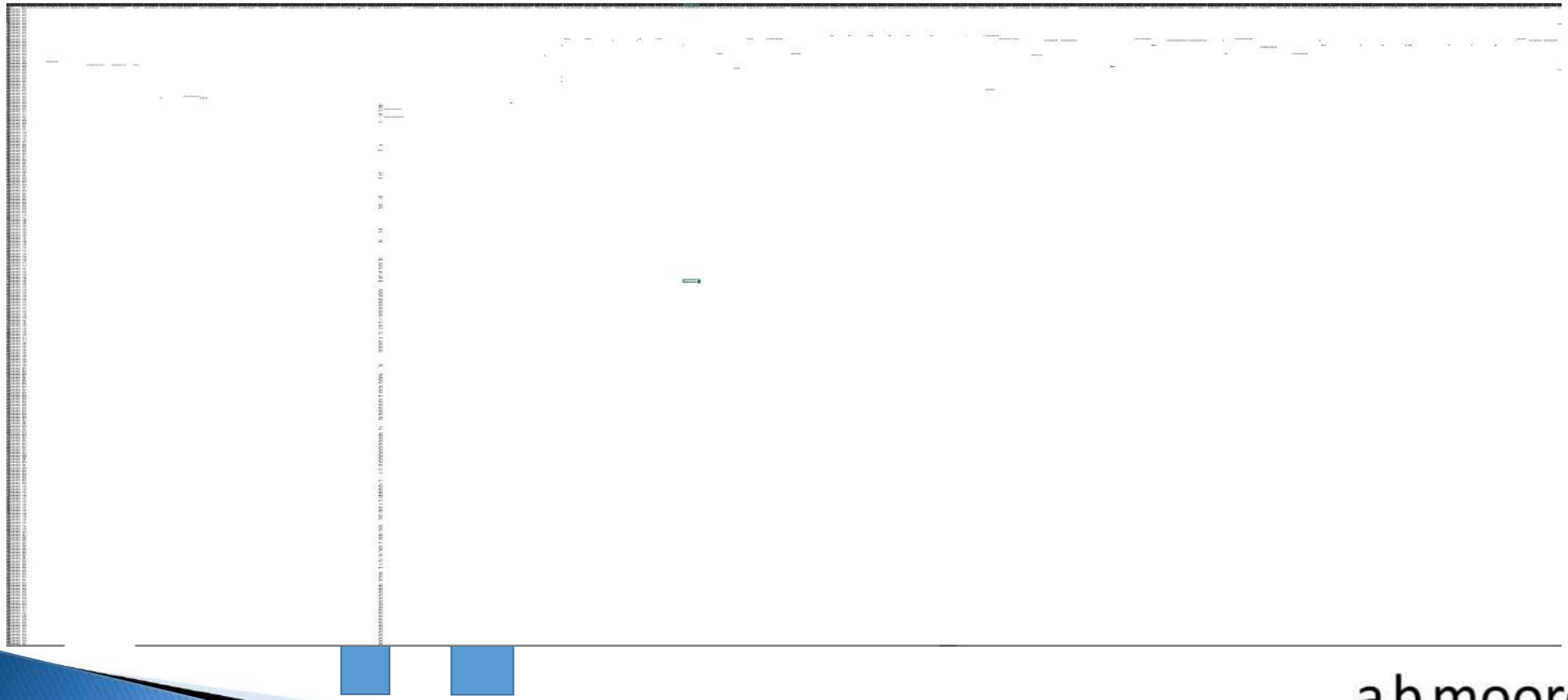
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Tesla-provided log data

VIN	DATE (US/Pacific)		VEHICLE SPEED (mph) (179.8 mph is recorded when vehicle is off)	ACCELERATOR PEDAL POSITION (%) (102 is recorded when vehicle is off)	STEERING ANGLE SENSOR (degrees) (positive indicates right turn, +/-819.1 is recorded when signal not available)	BRAKE APPLICATION	FRONT CRASH DETECTION
5YJSA1H1XEF	Aug-	:51:51.0 AM	15.3	40.8	1.6		
5YJSA1H1XEF	Aug-	:51:52.0 AM	17.6	0.0	1.9		
5YJSA1H1XEF	Aug-	:51:52.0 AM				ON	
5YJSA1H1XEF	Aug-	:51:53.0 AM	11.0	0.0	-15.8		
5YJSA1H1XEF	Aug-	:51:53.6 AM	6.1	0.0	-49.0		
5YJSA1H1XEF	Aug-	:51:54.0 AM	3.3	0.0	-56.5		
5YJSA1H1XEF	Aug-	:51:54.6 AM				OFF	
5YJSA1H1XEF	Aug-	:51:55.0 AM	0.2	0.0	-108.6		
5YJSA1H1XEF	Aug-	:51:56.0 AM	1.6	20.8	-126.1		
5YJSA1H1XEF	Aug-	:51:57.0 AM	4.9	31.6	-105.5		
5YJSA1H1XEF	Aug-	:51:57.5 AM	7.9	23.2	-43.5		
5YJSA1H1XEF	Aug-	:51:58.0 AM	9.8	23.6	13.3		
5YJSA1H1XEF	Aug-	:51:59.0 AM	12.0	39.2	114.3		
5YJSA1H1XEF	Aug-	:52:00.0 AM	17.7	41.6	35.7		
5YJSA1H1XEF	Aug-	:52:01.0 AM	24.8	48.8	-1.1		
5YJSA1H1XEF	Aug-	:52:02.0 AM	32.4	48.4	-3.0		
5YJSA1H1XEF	Aug-	:52:03.0 AM	39.8	49.2	-23.3		
5YJSA1H1XEF	Aug-	:52:04.0 AM	47.4	53.2	-31.8		
5YJSA1H1XEF	Aug-	:52:05.0 AM	55.3	55.6	-8.9		
5YJSA1H1XEF	Aug-	:52:06.0 AM	63.3	56.8	-6.7		
5YJSA1H1XEF	Aug-	:52:07.0 AM	70.6	64.8	-10.7		
5YJSA1H1XEF	Aug-	:52:08.0 AM	77.7	68.4	-6.2		
5YJSA1H1XEF	Aug-	:52:09.1 AM	84.5	70.8	-5.3		
5YJSA1H1XEF	Aug-	:52:10.0 AM	89.7	71.2	-4.8		
5YJSA1H1XEF	Aug-	:52:11.0 AM	94.7	70.8	-1.5		
5YJSA1H1XEF	Aug-	:52:12.0 AM	99.4	70.8	-1.0		
5YJSA1H1XEF	Aug-	:52:13.0 AM	103.4	70.4	-3.0		
5YJSA1H1XEF	Aug-	:52:14.0 AM	106.9	0.0	-3.4		
5YJSA1H1XEF	Aug-	:52:14.9 AM				ON	
5YJSA1H1XEF	Aug-	:52:15.0 AM	105.7	0.0	-3.7		
5YJSA1H1XEF	Aug-	:52:16.0 AM	97.1	0.0	-7.7		
5YJSA1H1XEF	Aug-	:52:16.6 AM					CRASH
5YJSA1H1XEF	Aug-	:52:16.6 AM	84.5	0.0	-13.9		
5YJSA1H1XEF	Aug-	:52:16.9 AM	78.3	0.0	-15.1		

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Tesla-provided owner-requested log data



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Tesla-provided owner-requested log data

Autosteer Hands On Detection	Autosteer State	Accelerator Pedal Position (%)	Brake Pedal Application	Cruise Control Set Speed	Cruise Control State	Gear Selection	Vehicle Speed (kph)
	AVAILABLE, 2.0						
		34.8					
							33.76
				22			
		35.2					
		35.6					
LC_HANDS_ON_REQD_NOT_DETECTED, 2.0							
				23			
		43.6					
				25			
LC_HANDS_ON_REQD_DETECTED, 1.0							
		43.2					
							42.16
				26			
		43.6					
	UNAVAILABLE, 1.0						
		40.8					
				28			

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Tesla-provided owner-requested log data

Autosteer Hands On Detection	Autosteer State	Accelerator Pedal Position (%)	Brake Pedal Application	Cruise Control Set Speed	Cruise Control State	Gear Selection	Vehicle Speed (kph)
		12.8					
					CRS_STATE_ENABLED, 2.0		
LC_HANDS_ON_REQD_NOT_DETECTED, 2.0							
	ACTIVE_NOMINAL, 3.0						
		12.4					
							109.04
		10.8					
		7.2					
		0					
							108.32

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Tesla-provided owner-requested log data

Selection of available parameters:

VIN
DATE (UTC)
Auto Lane Change State
Autosteer Hands On Detection
Autosteer State
Accelerator Pedal Position (%)
Brake Pedal Application
Cruise Control Set Speed
Cruise Control State
Vehicle Speed (kph)
Primary Steering Angle Sensor (degrees)
Primary Steering Column Torque Sensor (Nm)
Driver Brake Pedal Application
Estimated Brake Pedal Position
Lateral Acceleration (m/s^2)
Longitudinal Acceleration (m/s^2)
"Vehicle Yaw Rate.
(Positive indicates left turn)"
Wiper Speed
Hazard Warning Switch Status
Horn Switch Status

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Non-OEM Data Interpretation

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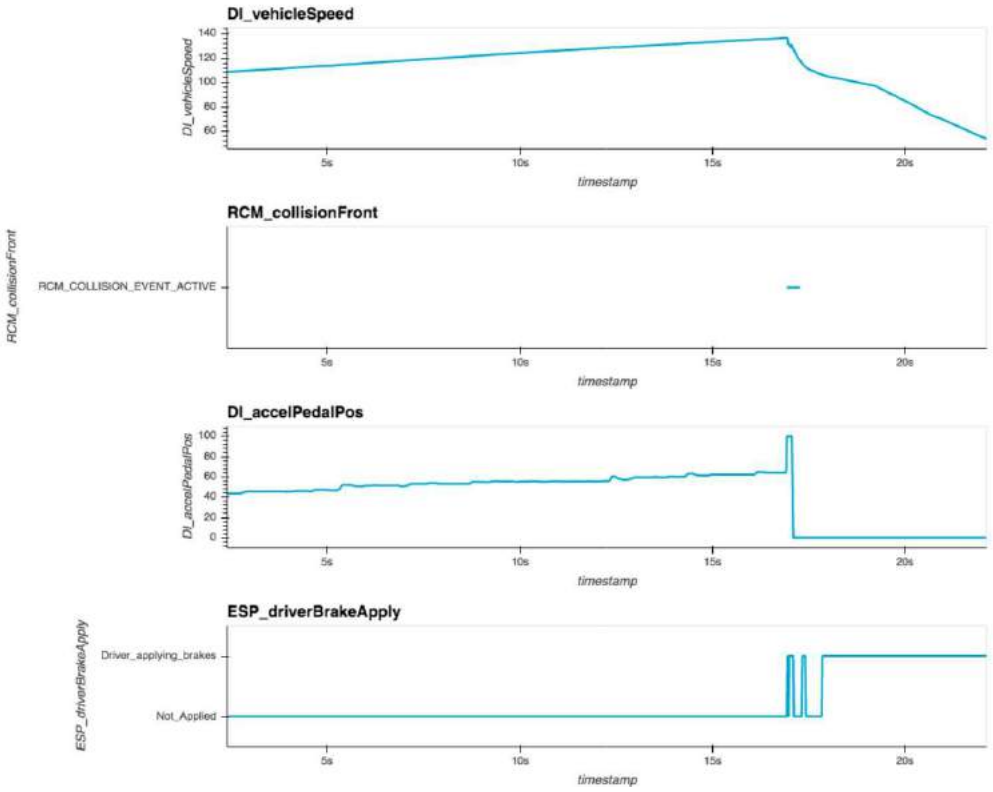
“Security Researchers” – Tesla crash



https://twitter.com/greentheonly/status/1326755485610020864

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“Security Researchers” – Tesla crash



https://twitter.com/greentheonly/status/1326755485610020864

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Tesla Log & Image Recording Summary

2014-2016 – on SD card behind main screen

2017 – SD card behind main screen or separate autopilot computer

2018-2020 - 3 sources

- Dashcam feature, using thumb drive in center console
- Video recorded and accessible by Tesla in a crash event
- SD card in AP computer, passenger footwell

2021+

- Same as above, but SD card logging currently unknown

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Tesla Log Data Summary

Gateway Logs – record at intervals and on changes, usually 1Hz continuously. Primary basis for data provided by Tesla in data requests.

“HRL” High Resolution Logs – recorded on specific triggers, including airbag deployment. 10-60 seconds of data per trigger, up to 50Hz.

CL logs – Software versions 2020 and newer. More difficult to reverse engineer.

Reverse engineering and evaluation of Tesla vehicle logs, F.C. Hoogendijk, MSc Netherlands Forensic Institute (NFI), 2020

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Tesla Autopilot Log Summary

AP1 – 2015-2016 – MobilEye EyeQ3 Hardware. Data transferred to gateway log over ~40 minute cycle after vehicle powers down.

AP2 & 2.5 – 2016-2018 – Nvidia Drive PX2 hardware with 8 cameras. Snapshot data recorded, with video. Deleted after successful upload to Tesla server.

AP3 – 2019+ - Tesla in-house “Full Self-Driving” hardware. Available data currently unknown.

“The encoding of these logs is proprietary and requires reverse engineering efforts to analyze.”

Reverse engineering and evaluation of Tesla vehicle logs, F.C. Hoogendijk, MSc Netherlands Forensic Institute (NFI), 2020

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GM Onstar Data From Research Study

"The LDW and FCA warnings are generated by the front camera module (FCM), which both use the same forward-looking camera. The FCM accumulates and stores into its memory a pre-determined set of data during an ignition cycle. The data elements are described in the next subsection and appendix. When an ignition-on event occurs, the previous ignition cycle's data are deleted. For the consenting vehicle owners in this study, OnStar remotely loaded a custom script onto the onboard OnStar module to retrieve the data from the FCM after ignition-off. This data is then sent over the air through the OnStar system during an ignition-off state." – *This applies only to vehicles in the subject study.*

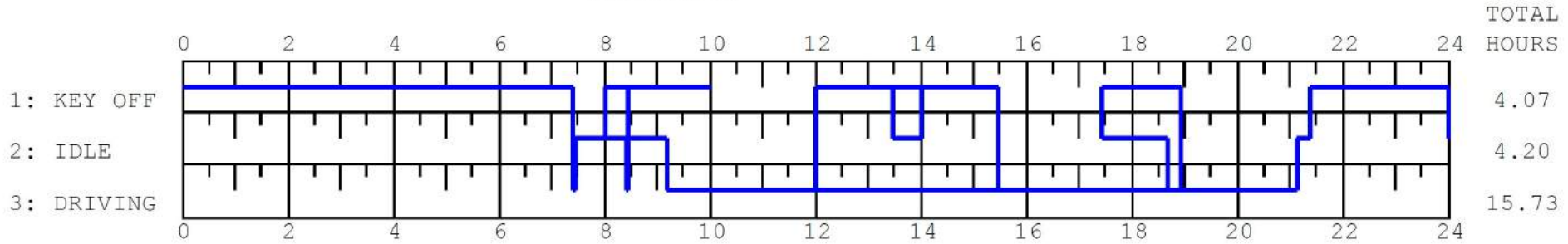
Large-Scale Field Test of Forward Collision Alert And Lane Departure Warning Systems, DOT HS 812 247, 2016
https://www.nhtsa.gov/sites/nhtsa.dot.gov/files/812247b-lanedeparturewarningsystems.pdf

EDR Inconsistencies – Risk of relying on OEMs for data interpretation

Date:	3/27/2018	
Start Time:	09:39:29	EST
Odometer:	6842.30	mi
Distance:	987.00	mi
Fuel:	127.25	gal
Fuel Economy:	7.76	mpg
Average Speed:	62.73	mph
DEF Fluid:	5.03	gal
Trip Economy:	196.36	mpg

Total (hh:mm)	15:44	04:12	04:04
Hour (EST)	Drive (min)	Idle (min)	Off (min)
00:00-02:00	0	0	120
02:00-04:00	0	0	120
04:00-06:00	0	0	120
06:00-08:00	3	34	83
08:00-10:00	2	24	94
10:00-12:00	170	70	-120
12:00-14:00	207	33	-120
14:00-16:00	240	0	-120
16:00-18:00	175	35	-90
18:00-20:00	80	40	0
20:00-22:00	67	15	38
22:00-24:00	0	1	119

(3/27/2018)



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Video Detectors at Traffic Signals



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NASS ADAS Reports; no related EDR files yet

Special Crash Investigations (Current)

New Search | New Filter Search

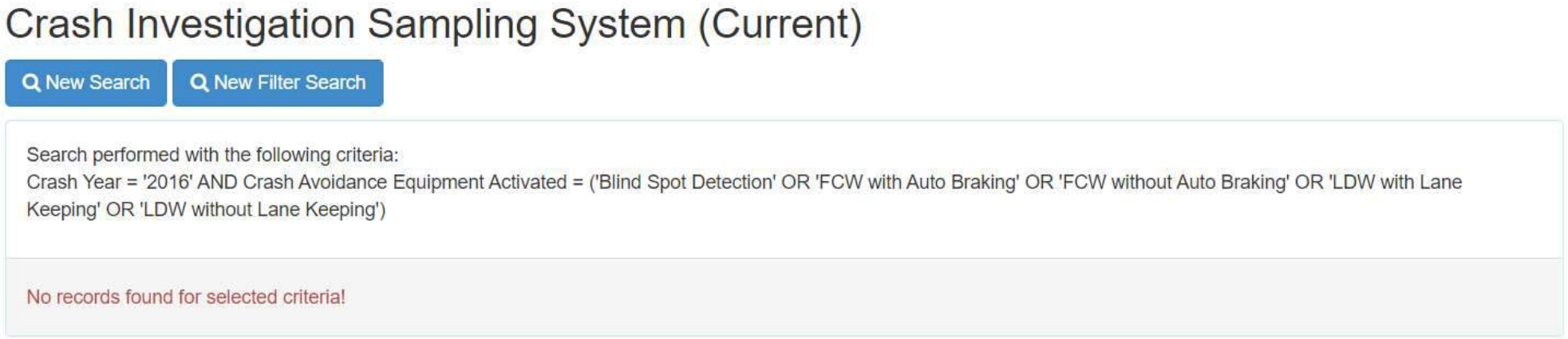
Search performed with the following criteria:
Crash Year = 'ALL' AND Crash Avoidance Equipment Activated = ('Blind Spot Detection' OR 'FCW with Auto Braking' OR 'FCW without Auto Braking' OR 'LDW with Lane Keeping' OR 'LDW without Lane Keeping')

Page: 1 ▾ Cases 1-2 of 2

View Case	Technical Report	Crash Date	SCI Case Type	State	Driver Age	Vehicle	Change In Velocity	Fatal
DS16027	DS16027	10/2016	Crash Avoidance Technology	AZ	20	2016 Honda Civic/CRX, del Sol	34 kmph	Non Fatal
DS17016	DS17016	7/2017	Rollover	CA	23	2017 Mercedes-Benz GLA Class	11 kmph	Non Fatal
View Case	Technical Report	Crash Date	SCI Case Type	State	Driver Age	Vehicle	Change In Velocity	Fatal

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CISS Crash viewer, 2016; no events with ADAS data available as of Jan 2020



Download CISS EDR events from the FTP Site:
https://www.nhtsa.gov/file-downloads?p=nhtsa/downloads/CISS/

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Electronic Data (EDR)

Summary:

- Toyota, Chrysler, Bendix, and Detroit Diesel have an abundance of ADAS-relevant data
- Although the data is useful, there are limitations
- Reverse engineering EDR data is time consuming, case specific, and not feasible for most forensic cases

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Pedestrian Automatic Emergency Braking (P-AEB)

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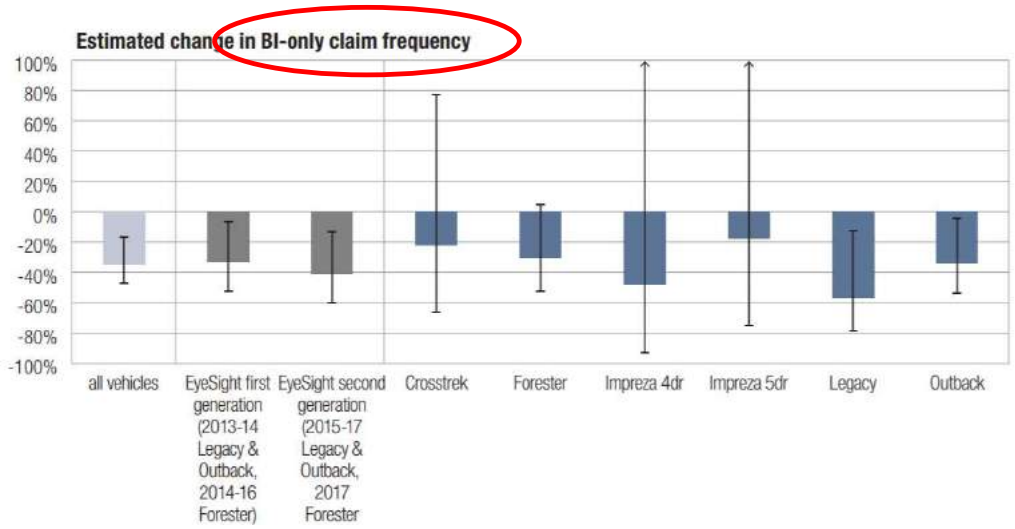
IIHS Blooper Reel



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Effectiveness of Subaru Pedestrian Detection

Subaru vehicles equipped with EyeSight reduced pedestrian-related claim frequency by 35 percent compared to Subaru vehicles without EyeSight



Effect of Subaru EyeSight on pedestrian-related bodily injury liability claim frequencies, Bulletin Vol. 34, No. 39 : December 2017
https://www.iihs.org/media/e8f617a5-b8f8-42e6-8b0f-9f56f9cb4208/W63Mlg/HLDI%20Research/Bulletins/hldi_bulletin_34.39.pdf

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What is P-AEB?

- Forward Collision Warning
- Brake Assist
- Automatic Emergency Braking
- Balancing risk of false positives and false negatives

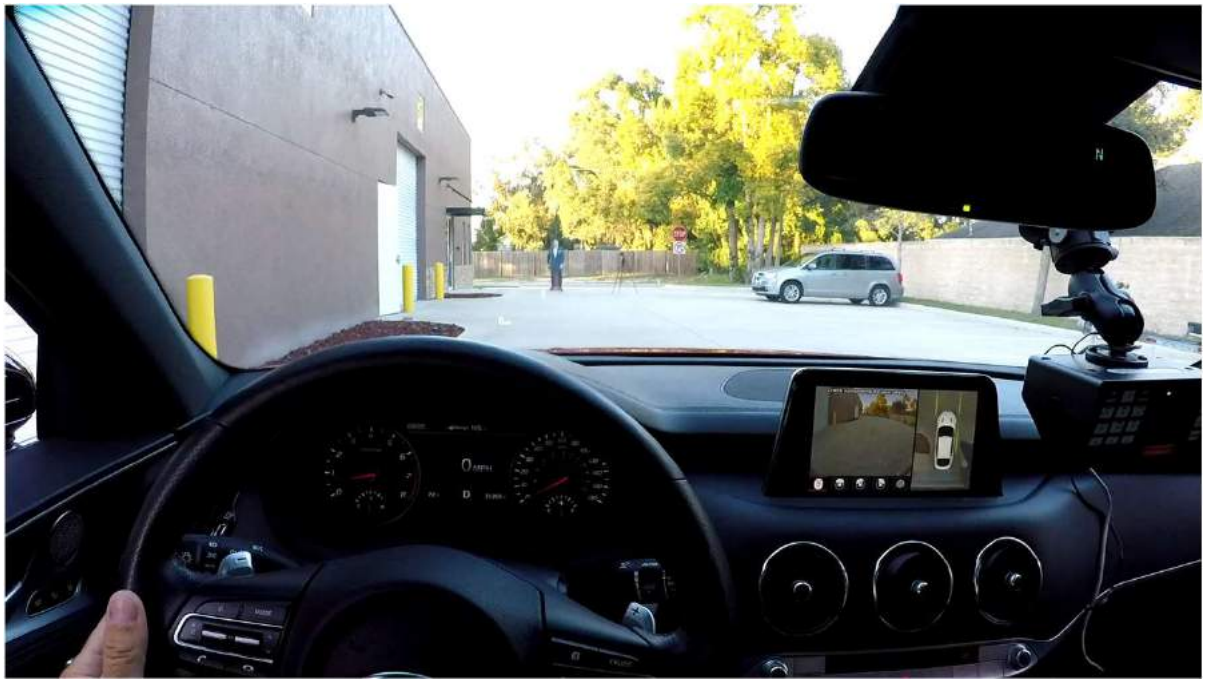
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2019 Ford Edge Pedestrian Detection – Camera Only



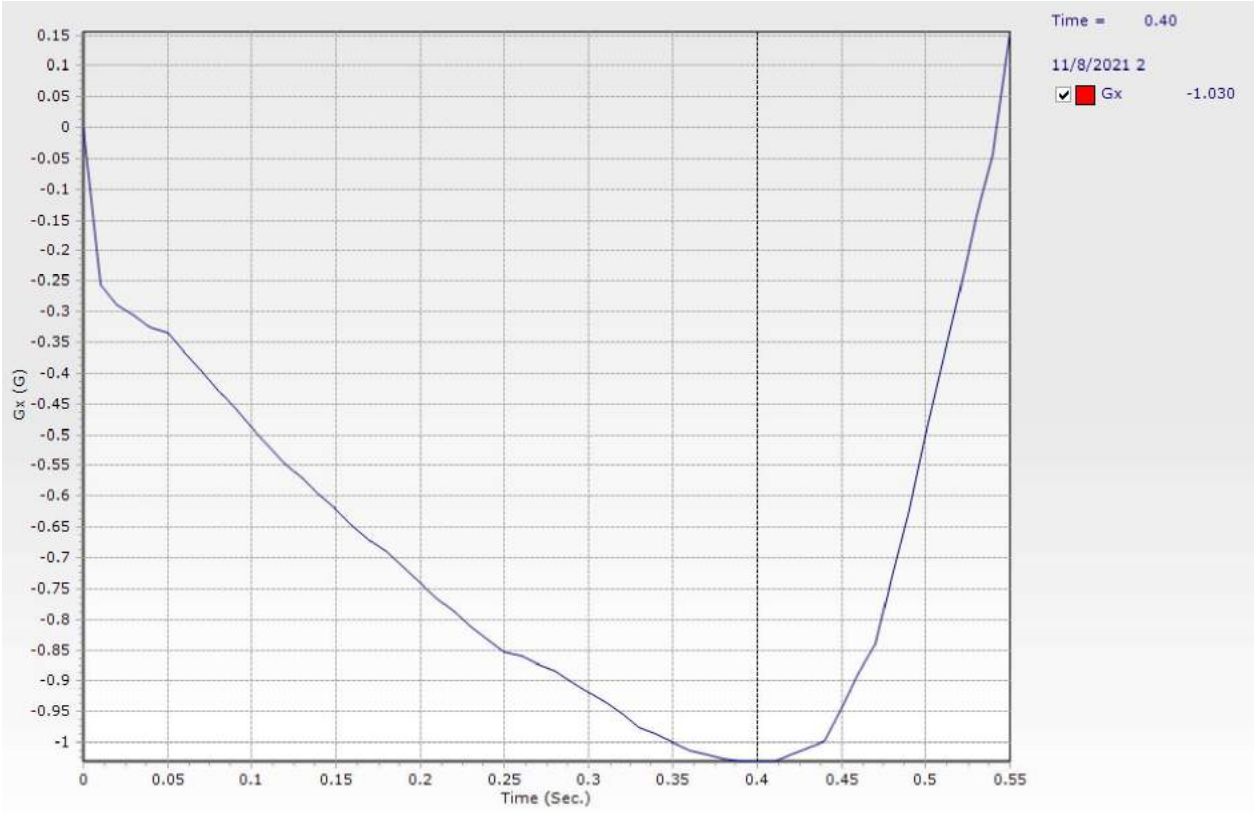
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2020 Kia Stinger vs James Bond



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2020 Kia Stinger vs James Bond



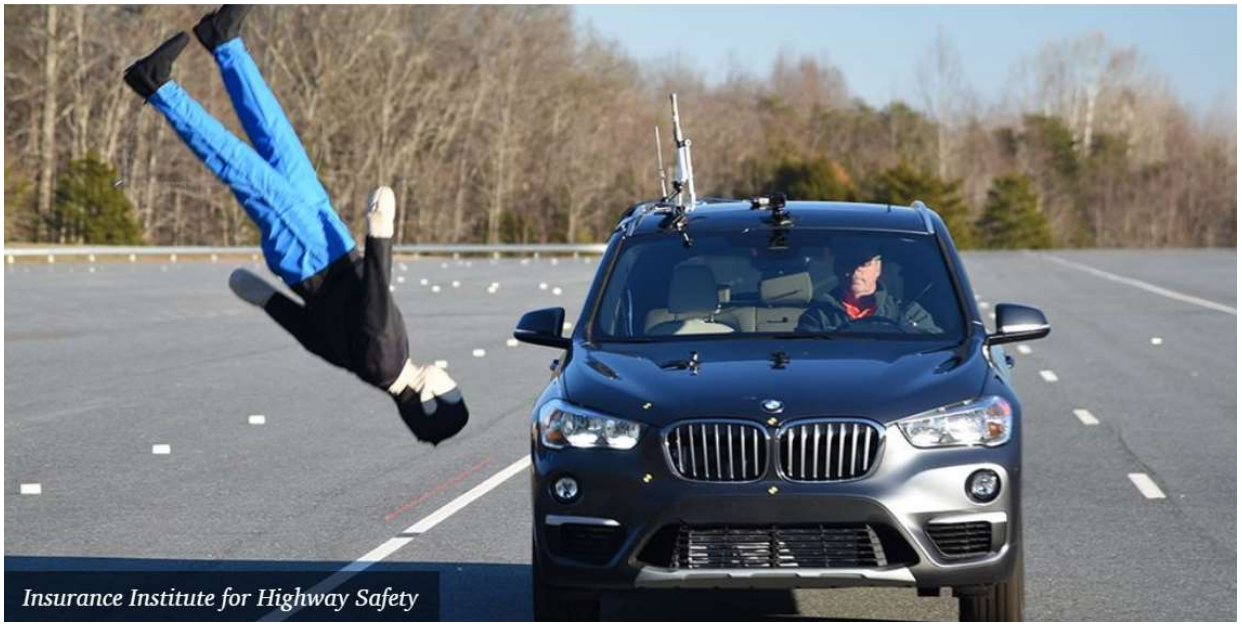
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2020 Kia Stinger vs Test Dummy



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Why is Pedestrian Detection Challenging?



- Short TTC
- Multiple trajectories – perpendicular, parallel, both moving, not in lane
- Sensing limitations
- Image classification
- Balancing risk
- Edge cases

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Why is Pedestrian Detection Challenging?



Edge cases exist, where the target's speed, gait, radar reflection, and image classification vary.

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Pedestrian Targets – 4activesystems

4activePS pedestrian static

4activeSystems produces dummies, which replicate human properties in size and shape. The child dummy represents a child at the age of 7 years. 4a dummies can be recognized by mono and stereo cameras as well as radar and infrared systems. In infrared is 50% refl ectivity, between 850nm and 910nm. There is no danger of crashes for the dummy, respectively the test vehicle, during a test.



- static EPT - Euro NCAP Pedestrian Target Static
- homogeneous distribution of radar cross section (RCS)
- visual signature - mono and stereo camera
- infrared signature (IR)
- crashable up to 60km/h
- adult weight < 4kg - child weight < 2kg

http://www.4activesystems.at/en/products/dummies/4activeps.html

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AAA Testing – Child darting out from between cars



https://www.youtube.com/watch?v=eEnklD0Vs4A

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Volvo P-AEB Failure at Car Dealership



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Test variables to watch for, in both published and exemplar testing

- Time of day and lighting
- Background objects and visual noise
- Driver inputs, which can inhibit systems
- Biofidelic motion
- Contrast
- Structure from motion

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P-AEB EDR - Toyota

Vehicle Control History via Techstream can provide:

- Target type
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response

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P-AEB EDR – General Motors

CDR 900 can provide:

- Some pedestrian classification
- Target speed toward vehicle, and across path
- Target distance
- Warnings and autobrake triggers
- Driver response
- Override/Inhibit Reasons

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Examples of ADAS – successes and the blooper reel

By the end of this module, you will:

- Recognize strengths and weaknesses of ADAS, and how it can influence a crash

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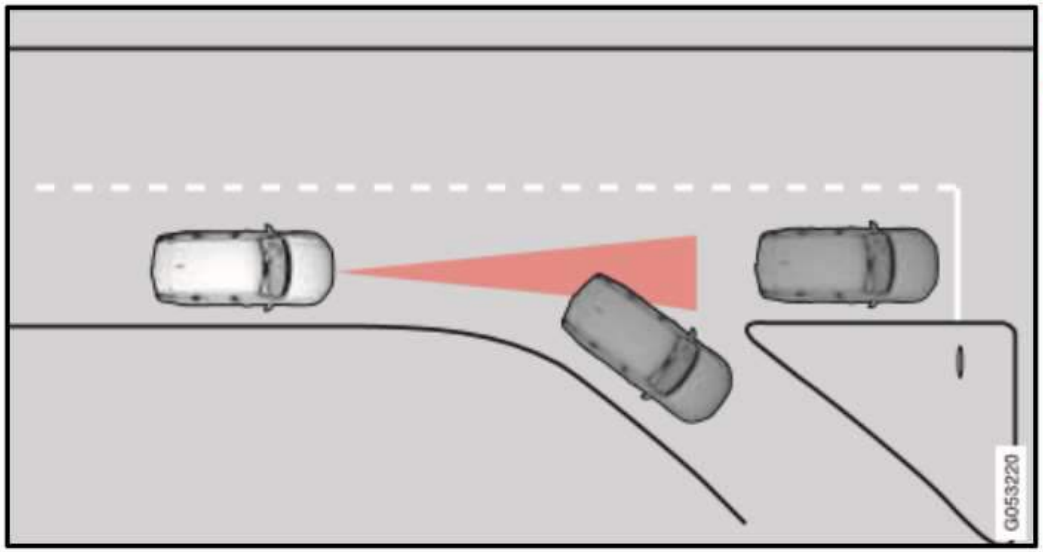
Examples of ADAS – Adaptive cruise

- 51st red light
- Cut-out scenarios
- Stopped vehicles with partial speed range ACC
- Stopped vehicles above ~50mph
- Radar tracking lead vehicle, but camera obscured by rain/snow

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Examples of ADAS – Adaptive cruise

- 2016 Volvo XC90 Owner's Manual



⚠ WARNING

If ACC changes targets from a moving vehicle to a stationary one at speeds **above** 20 mph (30 km/h), the system will **not** react to the stationary vehicle and will accelerate to the previously set speed.

The driver must actively apply the brakes to slow/stop the vehicle.

Tesla radar skip



Taking this one step further, a Tesla will also be able to bounce the radar signal under a vehicle in front – using the radar pulse signature and photon time of flight to distinguish the signal – and still brake even when trailing a car that is opaque to both vision and radar. The car in front might hit the UFO in dense fog, but the Tesla will not.

-Tesla blog post, late 2016

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Examples of ADAS – Automatic Emergency Braking

- Balancing risk of false positives and false negatives
  - Frontal vs rear impact probability
- Good performance only to classified objects
- Forward collision *mitigation*
- Distance used to alert driver by audible/visual/haptic warnings

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Examples of ADAS – Automatic Emergency Braking

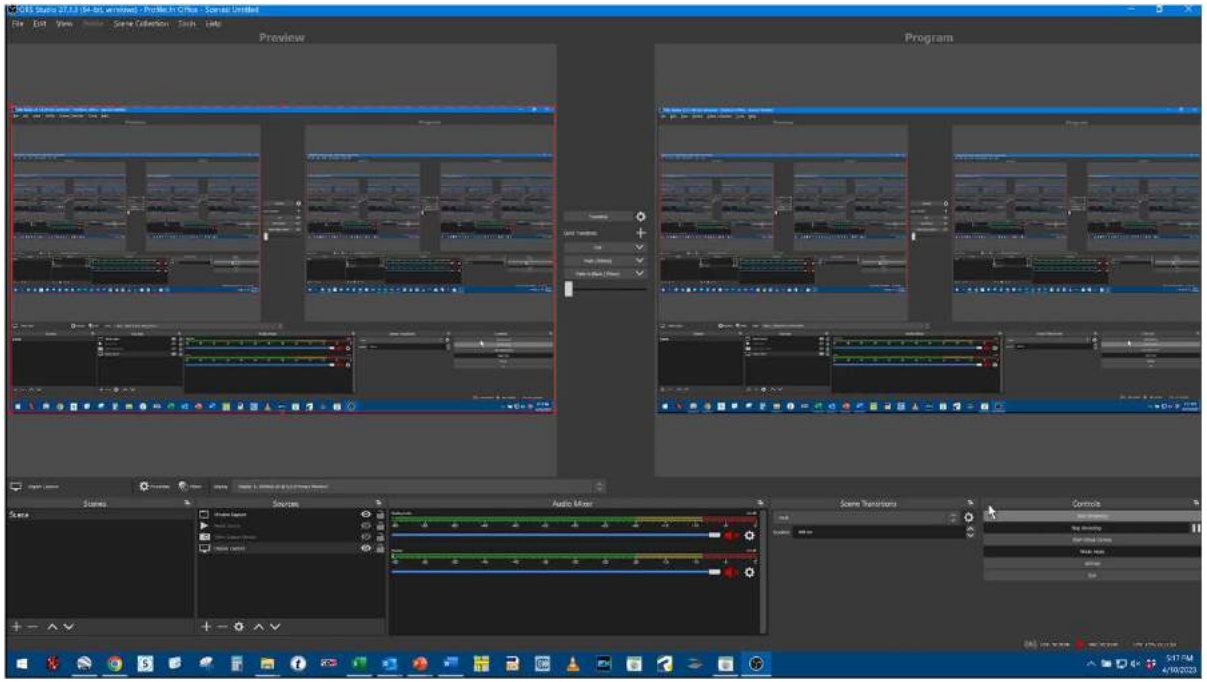


Ford F-150 recall due to false positives when passing a wide, reflective vehicle

Recall #15V614000, 09/30/2015, FORWARD COLLISION AVOIDANCE: ADAPTIVE CRUISE CONTROL

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Consequence of Unintended Braking



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Mazda Automatic Emergency Braking



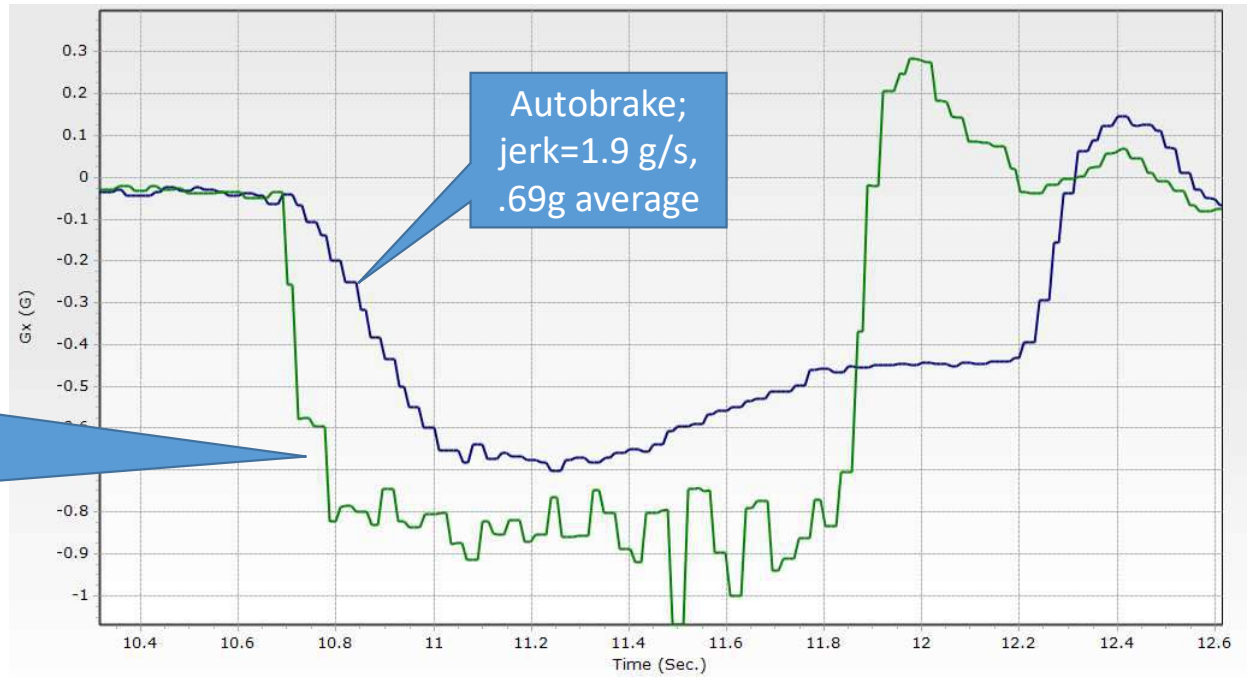
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2018 Toyota Camry Automatic Emergency Braking



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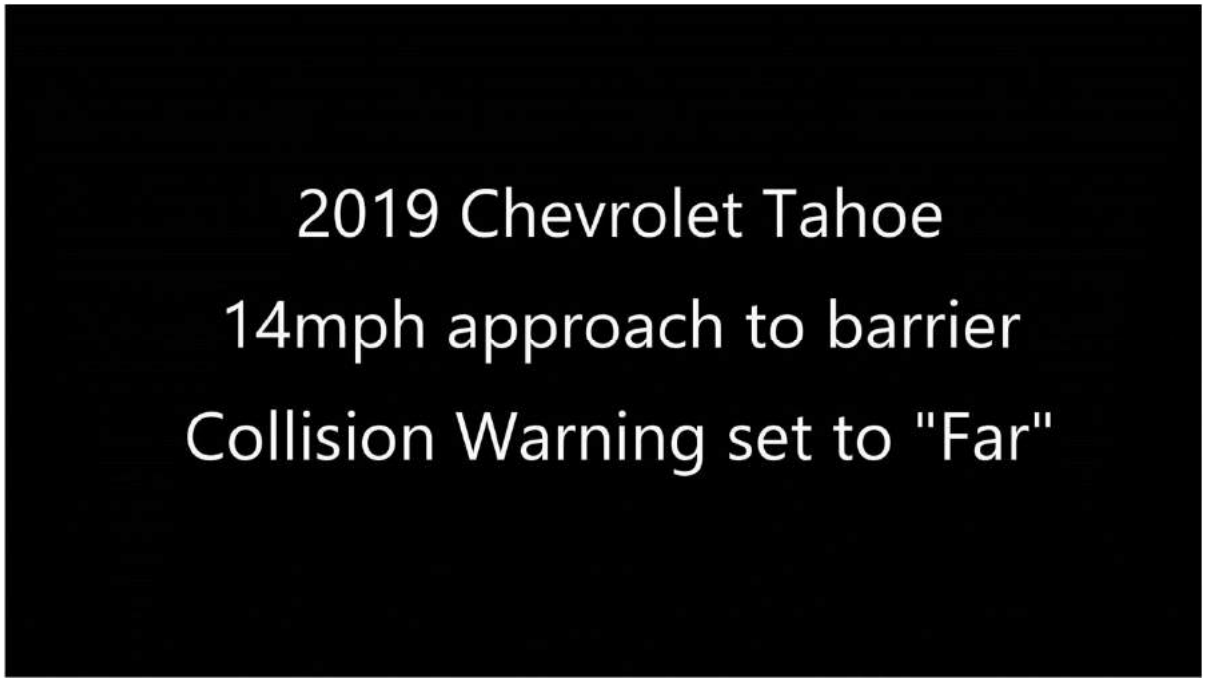
2018 Toyota Camry – Brake Assist v Autobrake



18mph. 0.9s from warning to autobrake

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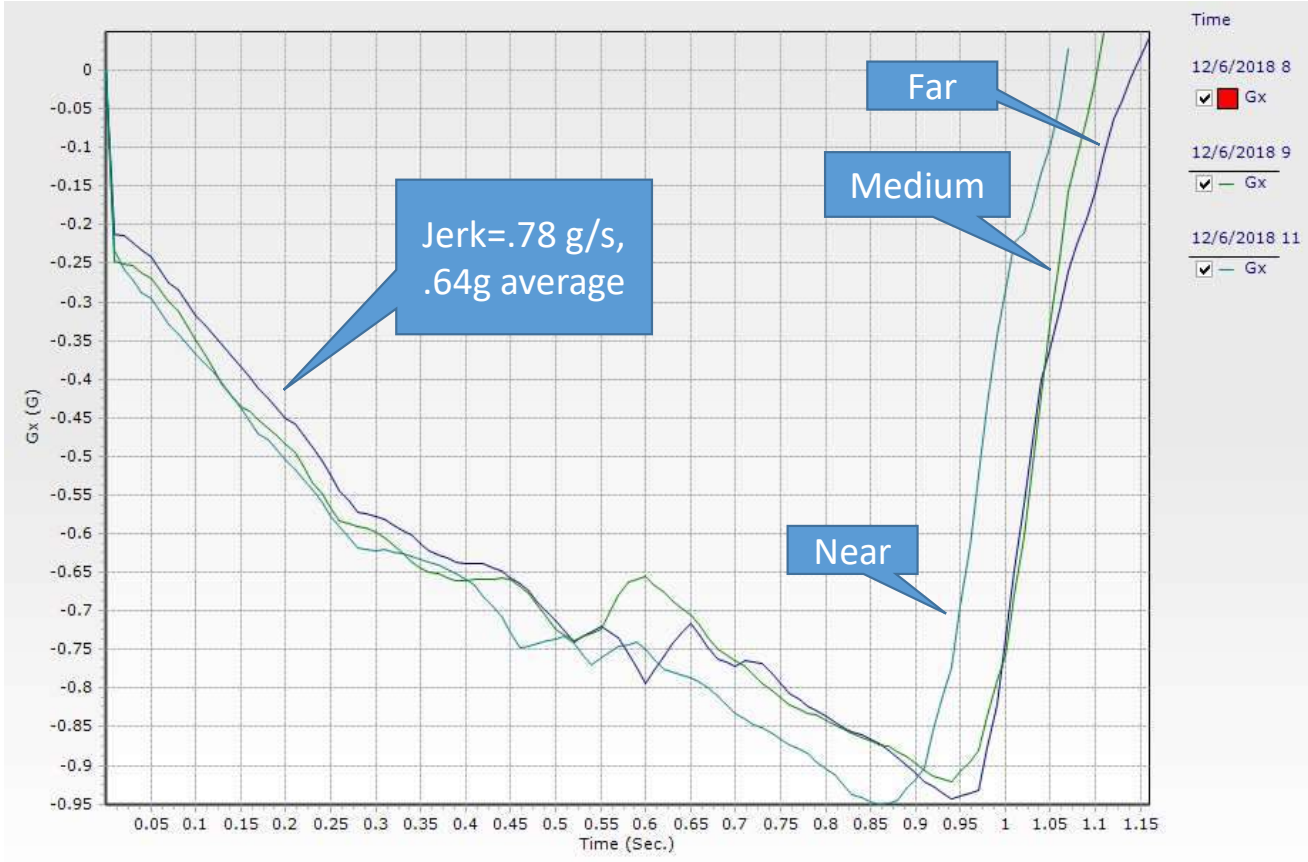
2019 Chevrolet Tahoe Automatic Emergency Braking



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2019 Chevrolet Tahoe Automatic Emergency Braking

Collision Warning Sensitivity Settings Indicated



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Tesla Collision Warning



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Examples of ADAS – Automatic Emergency Braking



Failure to classify and respond to an unusual vehicle

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Examples of ADAS – Automatic Emergency Braking



How did this happen?

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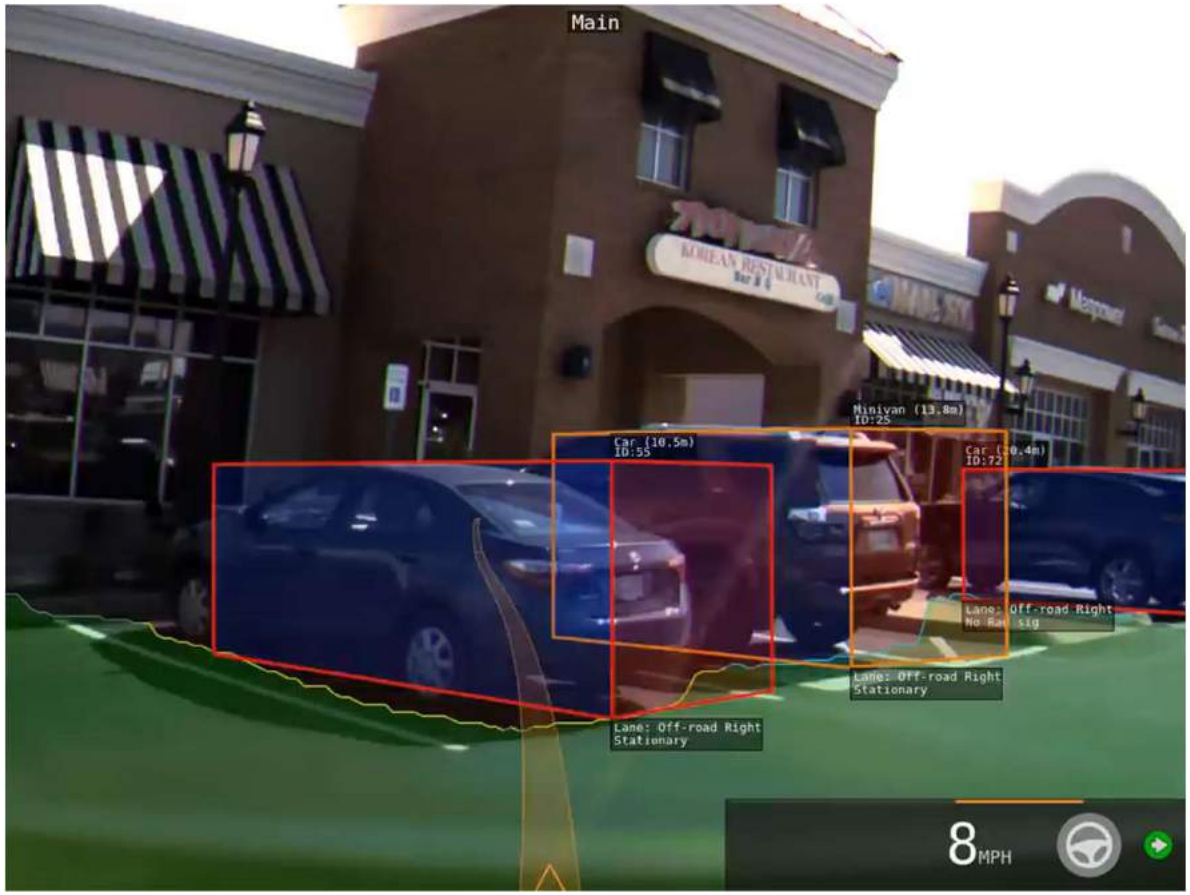
Tesla Emergency Vehicle Accidents as of 2021

First Responder Move-Over Incidents (Incident Date and location)

i.	2021-03-17	Lansing, MI
ii.	2021-02-27	Montgomery, TX
iii.	2020-08-26	Charlotte, NC
iv.	2020-07-14	Tucson, AZ
v.	2019-12-30	West Bridgewater, MA
vi.	2019-12-29	Cloverdale, IN
vii.	2019-12-07	Norwalk, CT
viii.	2018-05-11	South Jordan, UT
ix.	2018-05-29	Laguna Beach, CA

https://static.nhtsa.gov/odi/inv/2022/INIM-EA22002-87961P.pdf

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Tesla debug view of a firetruck – version 19.12
https://www.youtube.com/watch?v=9YBBc2eg27g

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Examples of ADAS – Automatic Emergency Braking

- Relevant to Adaptive cruise control (ACC):
- With a range of up to 160 meters and variable field of view, the MRR makes it possible to detect vehicles in front and vehicles merging at an early stage — making it the ideal basis for ACC systems. At speeds of up to 150 km/h (93 mph) and a maximum relative speed of up to 80 km/h (50 mph), …

Bosch MRR Radar Product Data Sheet

Examples of ADAS – Automatic Emergency Braking

- **The radar sensor has some performance limitations. If a stationary vehicle is in the vehicle's path, the AEB system will not function when the vehicle is driven at speeds over approximately 50 mph (80 km/h).**

2019 Nissan Sentra Owner's Manual and Maintenance Information, June 2019
https://owners.nissanusa.com/content/techpub/ManualsAndGuides/Sentra/2019/2019-Sentra-owner-manual.pdf

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Tesla Autopilot crash - Handan, China



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Las Vegas Autonomous Shuttle Crash



NTSB investigation HWY18FH001: https://www.ntsb.gov/investigations/AccidentReports/Reports/HAB1906.pdf

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Reverse Autobrake – Chrysler Pacifica



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Examples of ADAS – Lane Centering

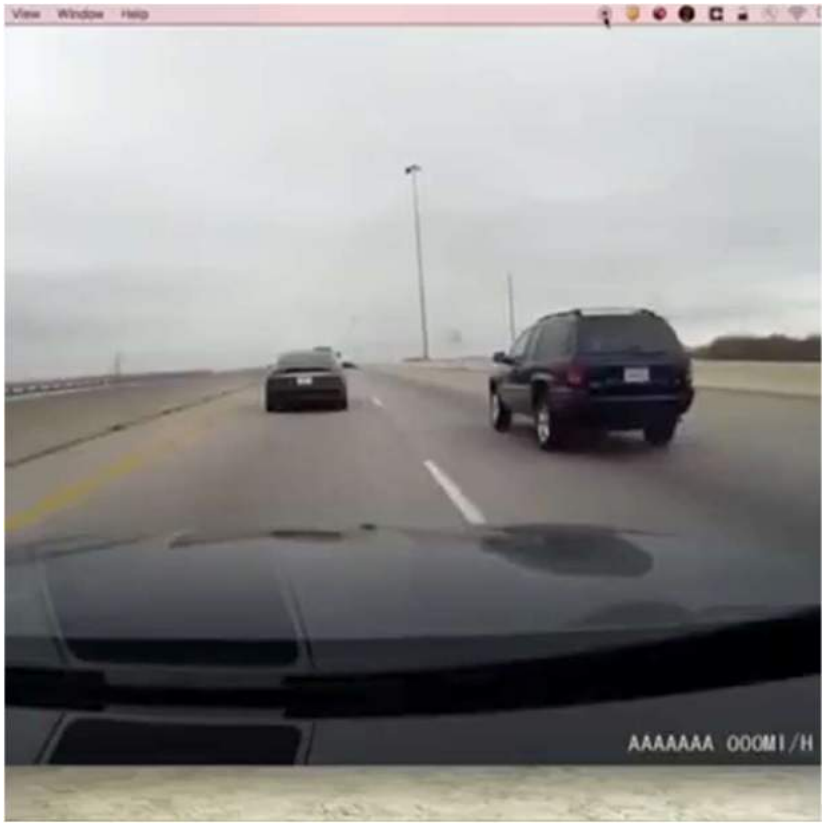
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Tesla S Construction Site



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Tesla autopilot barrier impact



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Tesla – Lane Line Confusion



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Tesla – Police Car Crash, Laguna Beach, CA



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Tesla – Police Car Crash, Laguna Beach, CA



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Tesla Impact with guardrail end treatment



NTSB Presentation on Mountain View, CA Crash – Crash Investigation HWY18FH011
https://www.ntsb.gov/news/events/Documents/2020-HWY18FH011-BMG-presentations.pdf#search=carlog

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Tesla Impact with guardrail end treatment

A



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Level 2/3 Autonomy* on a traffic circle



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Hyundai Ghost Ride



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Examples of ADAS – successes and the blooper reel

Summary:

- ADAS technologies have strengths and weaknesses
- ADAS can prevent a crash, but there are scenarios where it can contribute to a crash

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Media and Public Perception

By the end of this module, you will:

- Understand that public perception is dependent on the media, and is influenced by coverage of isolated events
- Be aware of road rage tendencies against autonomous vehicles

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Media and Public Perception

- Inflation of incidents
- Most drivers, when interviewed, claim they won't rely on ADAS technology.
- Studies show that drivers adopt and rely on ADAS tech too quickly and place too much trust[1].
- Using augmented reality to improve driver acceptance and Human-Machine Interface

[1] Waymo Safety Report, 2017

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Ethics – The Vulnerable Road User

- Ethics - “Self-driving cars programmed to decide who dies in a crash”
- The trolley problem is unlikely
- The most common ethics decision is brake vs not-brake
- But the VRU, specifically a child dart-out from between cars, is a unique case.

Self-driving cars programmed to decide who dies in a crash, 2017, Todd Spangler, Detroit Free Press

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Media and Public Perception

California public road "disengagements" by autonomous car prototypes
Dec. 2016 - Nov. 2017

Waymo	0.18 disengagements per 1K miles driven [largely suburban], 75 vehicles, 352,545 total miles
Cruise:	0.8 disengagements per 1K miles [urban driving, San Francisco], 90+ vehicles, 125,000 total mi
Nissan:	4.8 disengagements per 1K miles, 5 cars, 5,007 total miles
Zoox:	6.2 disengagements per 1K miles [primarily urban driving in San Francisco], 7 vehicles, 2,244 tc
Drive.ai:	15.4 disengagements per 1K miles, 7 vehicles, 6,015 total miles
Baidu:	24 disengagements per 1K miles, 4 cars, 1,971 total miles
Telenav:	31.3 disengagements per 1K miles [highway primarily], 1 vehicle, 1,824 total miles
Aptiv/Delphi:	45.3 disengagements per 1K miles [mainly highway miles], 1 car, 1810.6 total miles
Nvidia:	215.8 disengagements per 1K miles, 2 cars, 505 total miles
Bosch:	412 disengagements per 1K miles [almost entirely highway miles], 3 cars, 1,454 total miles
Valeo:	375 disengagements per 1K miles [curved sections of highways], 1 vehicle, 574 total miles
Mercedes:	774.6 disengagements per 1K miles, 3 vehicles, 1,087 total miles

5,555 miles per driver intervention

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Media and Public Perception

By the end of this module, you will:

- Understand that public perception is dependent on the media, and is influenced by coverage of isolated events
- Be aware of road rage tendencies against autonomous vehicles

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Looking into the future

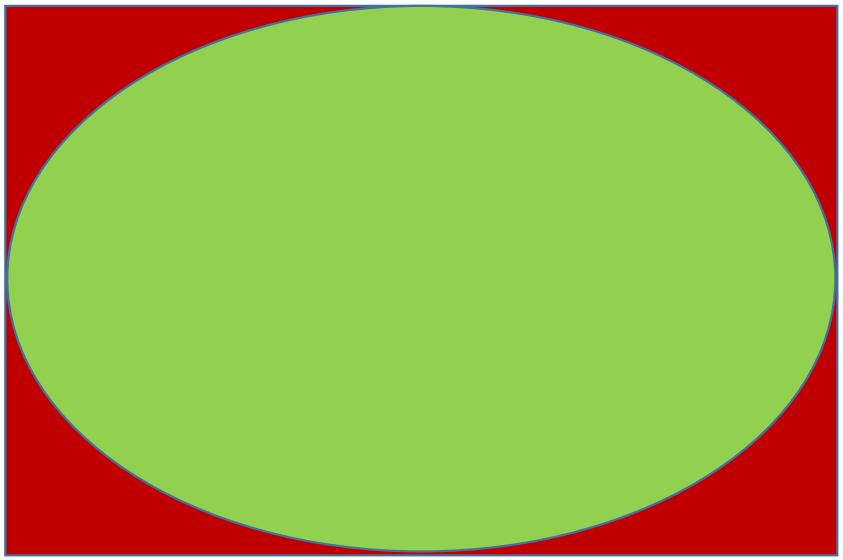
By the end of this module, you will understand the significance of :

- Edge cases
- Fallback behaviors
- Limiting Operational Design Domains (ODD)

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Looking into the future – edge cases

- Unusual and unpredictable scenarios
- Fallback behaviors
- Limiting Operational Design Domains (ODD)



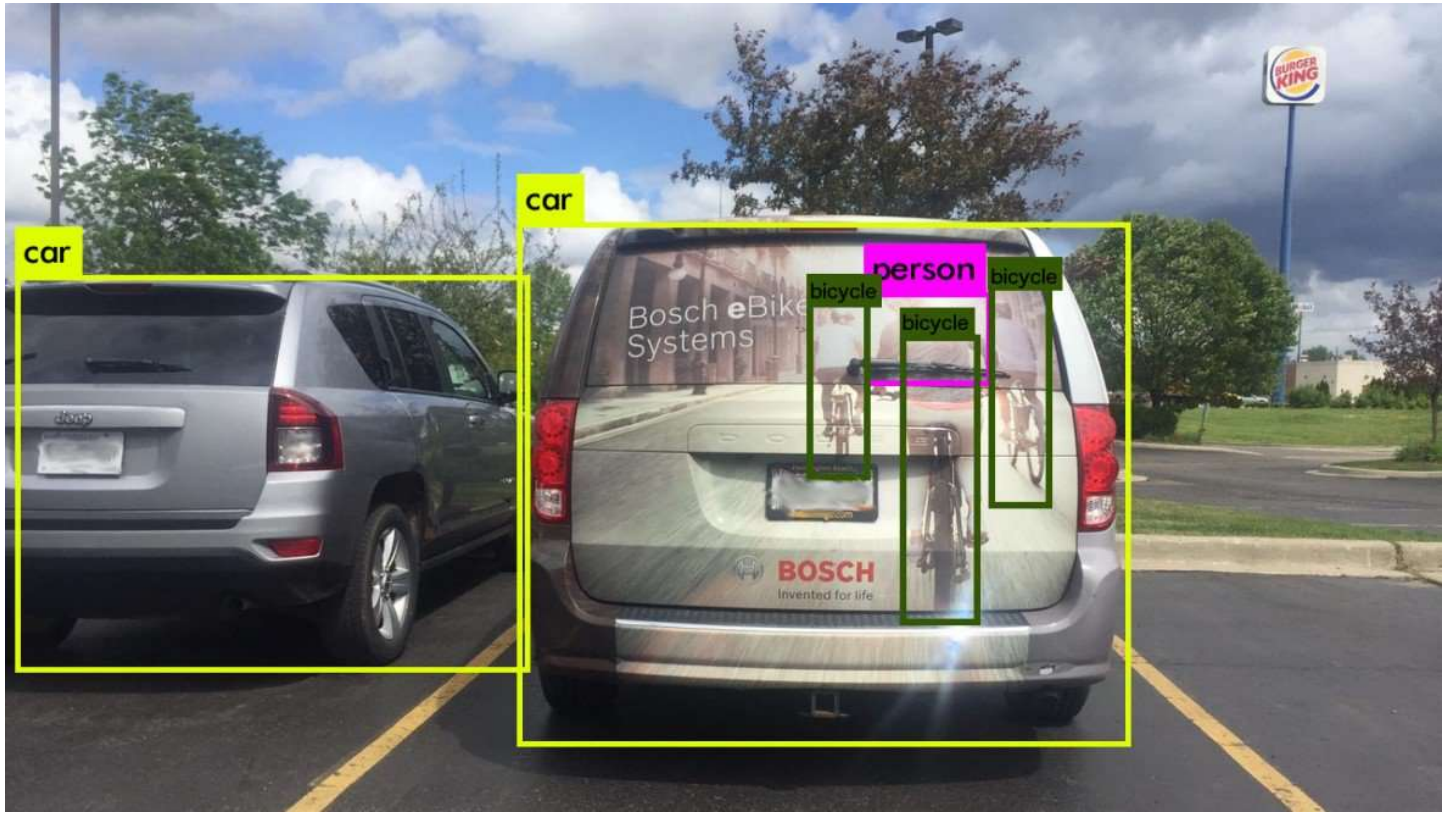
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Edge case – wheelchair chasing a chicken



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Edge case - Machine vision inadvertent image classification



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Looking into the future

Summary: The future of ADAS accident reconstruction will involve developments in:

- Edge cases
- Fallback behaviors
- Limiting Operational Design Domains (ODD)

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Further Learning

- SAE Reconstruction of Autonomous Vehicle Accidents
- SAE Introduction to Highly Automated Vehicles C1603
- SAE ADAS Application: Automatic Emergency Braking C1704
- Udacity.com – Self-Driving Cars “Nanodegree”
- MIT’s Artificial Intelligence Lectures – Lex Fridman on Youtube
- Innovators’ blogs on Medium.com – Sebastian Thrun, comma.ai, Andrew Ng
- Good List of publications: http://www.cvlibs.net/projects/autonomous_vision_survey/

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Alan Moore, P.E.
A.B.Moore Forensic Engineering, Inc.
Orlando, FL
alan@abmoore.com
321-946-1283

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Wrap up and Questions

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Acronym	Description of Acronym	Definition
5G	Fifth-generation mobile network	Upcoming telecommunications standard expected to deliver 10 Gbps and ultra-low latency.
ACC	Adaptive Cruise Control	Cruise control system that automatically adapts speed to maintain safe distance from vehicles in front
ADAS	Advanced driver assistance system	Vehicle systems designed to improve driving safety
AEB	Automatic Emergency Braking	System that detects an impending forward crash with another vehicle in time to avoid or mitigate the crash.
AI	Artificial Intelligence	Intelligence and decision-making that come from a machine
AV	Autonomous Vehicle	Vehicle that is capable of driving itself, typically classified at levels 3, 4, or 5 for driving automation
BA	Brake Assist	Helps in detection of attempted panic braking based on the force that is applied to the brake pedal and how fast the driver is stepping on the pedal
BSM	Blind Spot Monitoring	ADAS applications that monitors the driver's blind spots at the rear quarters of the car and provides visual, audible, and/or tactile alerts when a vehicle is present in a situation requiring such alerts
BSM	Basic Safety Messages	DSRC data broadcasted from vehicles through V2V and V2I at a frequency of 10Hz. Core contents are data elements that describe a vehicle's position (latitude, longitude, elevation) and motion (heading, speed, and acceleration)
CAMP	Collision Avoidance Metrics Partnership	Partnership formed by vehicle OEMs to accelerate the implementation of crash avoidance countermeasures in passenger vehicles to improve traffic safety

CIB	Crash Imminent Braking	Systems automatically apply the brakes in a crash imminent situation if the driver does not respond to warnings
DBS	Dynamic Brake Support	Supplements a driver's commanded brake input by increasing the output of the foundation brake system, for the purpose of mitigating the severity of, or , in some cases avoiding , rear-end collisions
DDT	Dynamic Driving Task	Various activities required to operate a moving vehicle in an environment that changes due to alignment, weather, roadside conditions, etc. Tasks include steering, braking, and accelerating.
Deep Learning	Deep Learning	Subset of machine learning that involves many layers of processing, massive amount of data, and enormous computing capacity. These algorithms can facilitate computer vision, natural language processing, driving strategy, personalization, and decision-making.
DSRC	Dedicated Short Range Communication	Communications protocol developed to address the safety critical issues associated with sending and receiving data among vehicles and between moving vehicles and fixed roadside access points.
FCW	Forward Collision Warning	System comprising of sensors, computational capability, software and alert transducers which function to warn the driver of potential forward collision
Fuse	Sensor Fuse	Second of the 3 stages of in-vehicle compute required for autonomous driving (sense, fuse, decide). The vehicle correlates and fuses sensor data to create a model of its environment.

Haptic Alert	Haptic Alert	Warning modality in which a transducer produces a tactile sensation to the driver. Implemented as vibration in either the driver's seat pan or the steering wheel.
HAV	Highly Automated Vehicles	SAE Levels 3-5 vehicles with automated systems that are responsible for monitoring the driving environment, in order to draw a distinction between Levels 0-2 and 3-5 based on whether the human operator or the automated system is primarily responsible for monitoring the driving environment
Headway	Headway	Longitudinal distance from the front most location of the SV to rear most point on the POV or to the fixed point on the ground longitudinally coincident with the rear most point on a stationary POV.
HMI	Human-machine Interface	Interface responsible for two-way communication between a vehicle and its occupants. Touchscreen displays, voice recognition, or integration with mobile devices.
IoT	Internet of Things	Network of physical objects - devices, vehicles, buildings, machines, and other items - embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data.
LCA	Lane Centering Assist	Combines a forward-facing camera to detect lane markings with an electric steering system, keeping the vehicle in the center of the lane.
LDW	Lane Departure Warning	Uses a forward-facing camera to detect lane markings, warning the driver in case the vehicle leaves the lane without proper use of the turn signal.
LIDAR	Light Detection and Ranging	Laser based session of Radar and used in autonomous vehicles. Sits on top of the vehicle roof to enable unhindered 360-degree view of the area surrounding the vehicle. 64 lasers spin at about 900 rpm and create a detailed 360-degree 3D map of the surrounding environment in order to view all obstacles in real time.
LKA	Lane Keeping Assist	Combines a forward-facing camera to detect lane markings with an electric steering system, keeping the vehicle in the center of the lane.
LV	Lead Vehicle	Principal Other Vehicle (POV) when the potential collision is a Rear-end collision with the POV.
Machine learning	Machine Learning	Subset of AI that gives machines the ability to learn on their own, resulting in algorithms that make data-driven decisions.

MV	Machine Vision	Machine-assisted moderation of images and the ability to extract rich information from images to categorize and process visual data.
ODD	Operational Design Domain	Domain describing the specific conditions under which a given automated driving system or feature is intended to function, in the automated driving environment. Exp: Freeways, downtown areas, roadways with speed limits <35 mph.
OEDR	Object and Event Detection and Response	Detection by the driver or automated driving systems of any circumstance that is relevant to the immediate driving task, as well as the implementation of the appropriate driver or system response to such circumstance.
OEM	Original Equipment Manufacturer	
OTA	Over-the-air Updates	Software or firmware updates to a vehicle that are downloaded from the cloud.
PAEB	Pedestrian Automatic Emergency Braking	System that performs automatic braking in cases a pedestrian is detected in front of the vehicle.
Park Assist	Park Assist	System of ultrasonic sensors on the front and/or rear bumpers that provide the driver with visual, audible, and/or tactile alerts as their vehicle approaches a stationary object.
Platooning	Platooning	Method of grouping vehicles together, primarily to increase the road capacity, which will lead up to fully automated vehicles on a wide scale. Will decrease the distances between cars or trucks and would allow vehicles to accelerate or brake simultaneously.
Rain-Sensing Wipers	Rain-Sensing Wipers	Windshield wipers whose rate of operation is electronically controlled based on the amount of moisture on the windshield.
RCTA	Rear Cross-Traffic Alert	Systems let you know if you're about to back into oncoming cross traffic. Multiple sensors or wide angles cameras are located near the rear of the vehicle, detecting traffic that comes from the side, typical parking lot situations.
RTK	Real Time Kinematics	Technique is based on the use of carrier measurements and the transmission of corrections from the base station, whose location is well known, to the rover, so that the main errors that drive the stand-along positioning cancel out.
SPaT	Signal Phase and Timing	Signal state of the intersection and how long this state will persist for each approach and lane that is active.



SPaT	Signal Phase and Timing	Signal state of the intersection and how long this state will persist for each approach and lane that is active.
TTC	Time to Collision	Time interval, usually measured in seconds, required for one vehicle to strike another object.
V2X	Vehicle-to-everything	Communication between a vehicle and the cloud, other cars (vehicle-to-vehicle or V2V) and infrastructure (vehicle-to-infrastructure, V2I)
PIB	Post-Impact Automatic Braking	Decelerates the vehicle after a collision occurs.